

When Common Time Trends Are Not Parallel. Double Selection in Causal Identification*

Christian Alemán
New York University Abu Dhabi

Christopher Busch
LMU Munich
CESifo and CEPR

Raül Santaeulàlia-Llopis
New York University Abu Dhabi
UAB, BSE and CEPR

Tetyana Surovtseva
Universitat de Barcelona
IEB

August 13, 2025

PRELIMINARY AND INCOMPLETE

Abstract

In Difference-in-Difference designs, typically a choice is made between considering a formulation in the level or the logarithm of the outcome. This amounts to choosing a data generating process, which allows for either group-specific level or proportional differences. The assumption of a constant selection bias (CSB) between groups then translates into an assumption of parallel time trends in either levels or logs. We introduce a framework that simultaneously allows for constant group-specific level and proportional shifters of a common time trend (CTT). In this framework of constant *double selection* (DS), a CTT does not manifest itself as a parallel trend. The induced selection bias is a linear combination of the one arising under CSB in levels and the one arising under CSB in logs. Both of these are nested as special cases. While the selection bias is not constant over time under DS, the expected outcome of one group is a linear transformation of the expected outcome of the other. This allows to uniquely identify double selection in the outcome-on-outcome space, and to identify causal effects of binary policy treatment in its presence. We propose an estimator that operates in levels and conduct inference. We illustrate the relevance and versatility of our framework by revisiting [Card \(1992\)](#): Allowing for double selection doubles the number of possible states that can deliver a credible counterfactual for California in the context of an increase of the minimum wage.

Keywords: Logs, Levels, Causal, Identification, Difference-in-Difference, Double Selection

JEL Classification: C01, H00, E01, E22, E25

*We thank participants at seminars and conferences for helpful comments and suggestions. Raül Santaeulàlia-Llopis acknowledges financial support from the AGAUR 2020PANDE00036 "Pandemies" Grant 2020-2022, from the Spanish Ministry of Economy and Competitiveness, through the Proyectos I+D+i 2019 Retos Investigación PID2019-110684RB-I00 Grant, Europa Excelencia EUR2021-122011 and the Severo Ochoa Programme for Centres of Excellence in R&D (CEX2019-000915-S).

Alemán: christian.c.aleman@gmail.com; Busch: chris.busch.econ@gmail.com; Santaeulàlia-Llopis: lo-raulet@gmail.com; Surovtseva: tsurovtseva@ub.edu

"If there is a common trend in logs, there will not be one in levels and vice versa"
(Angrist and Pischke, 2009, p. 171).

1 Introduction

Difference-in-differences (DiD) is a workhorse method in empirical economics for identifying causal effects of policy interventions or natural experiments. Its appeal lies in its transparency and simplicity: identification hinges on comparing changes over time between the treated and the control groups under the premise that, absent treatment, these groups would have followed parallel trends. When this condition holds, the DiD estimator delivers credible estimates of the *average treatment effect on the treated*, even in the presence of selection bias on unobservables inherent in observational data.¹ At the core of this identification strategy lies the assumption that, in the absence of treatment, any systematic differences between the groups would have remained unchanged over time: in other words, DiD assumes a constant selection bias (CSB).²

The validity of the constant selection bias assumption hinges critically on the scale in which outcomes are measured. When outcomes are expressed in levels, the DiD estimator assumes a constant *additive* difference between the treated and the control groups, whereas when expressed in logs, DiD assumes a constant *proportional* difference. However, these two assumptions cannot hold simultaneously: if the selection bias is constant in levels, then it typically varies in logs, and vice versa.³ As noted by Angrist and Pischke (2009) in the leading quote, this trade-off is inherent to the structure of DiD. The same issue arises with any other monotonic transformation of the outcome variable. Thus, the choice between levels and logs implicitly imposes assumptions about the data generating process (DGP), and can significantly impact causal inferences drawn (e.g. Meyer, 1995; Lechner et al., 2016; Kahn-Lang and Lang, 2020; McConnell, 2024).

In this paper, we introduce a unified framework that encompasses both the levels and logs specifications of DiD and establishes identification of treatment effects in settings that violate the CSB assumption in either of the two traditional variants of DiD. Rather than transforming the outcome variable, our approach acknowledges that the underlying process may involve both additive and proportional selection biases simultaneously, which we refer to as *double selection*. To this end, we propose a DGP featuring a common time trend (CTT) and two types of group-specific shifters: a level shifter (additive to the CTT) and a proportional shifter (multiplying the

¹The popularity of DiD is further reinforced by its straightforward implementation via an OLS regression.

²Combined with the Stable Unit Treatment Value Assumption (SUTVA) and the absence of Pre-Treatment Effects (NPE), a constant selection bias (CSB) delivers identification of the effect of treatment.

³Both are constant simultaneously only if outcomes are identical and the selection bias is zero in both levels and logs.

CTT). In the presence of double selection, the *common* time trend does not manifest as *parallel* time trends: the difference between the groups is not constant over time, whether the outcome is expressed in levels or in logs. The presence of only one of the two types of selection (and thus a setting with parallel time trends) is captured as a constrained subcase of this more general framework.

We then formalize this framework by assuming a *Constant Double Selection Bias* (CSB), which, together with the standard Stable Unit Treatment Value Assumption (SUTVA) and the assumption of No Pre-Treatment Effects (NPE), ensures identification of treatment effects. Notably, in the presence of double selection, common pre-trends remain undetectable when considering group level outcomes over time, rendering variants of the standard DiD approach inapplicable. Consider the selection bias in levels.⁴ In any given period, it can be expressed as a linear combination of two components: a constant term, which captures the level shifter, and a time-varying term, which captures the proportional shifter. We thus recast the model in an outcome-on-outcome representation. A constant difference between two groups in both the level and the proportional shifters implies that the expected outcome of one group is a linear transformation of the expected outcome of the other. Importantly, this relationship does not hinge on any restrictions regarding the CTT itself, which can take on the form of a deterministic trend component, or include a stochastic component.

Guided by this insight, we develop a two-step estimation procedure that leverages the outcome-on-outcome relationship implied by constant double selection. Our approach yields estimates of the treatment effect under minimal data requirements: relative to the standard DiD, it requires only one additional pre-treatment period. In the first step, we use pre-treatment data to estimate a linear projection of the average outcomes of the control group on those of the treated group. The two coefficients from this projection, which can be implemented by a simple OLS regression, capture the selection in levels and in logs (as determined by the different level and proportional shifters of the two groups). In the outcome-on-outcome space, this linear relationship under double selection also allows for a simple visual inspection of the data when plotting the outcome of the control group against the outcome of the treatment group. In the second step, we use these estimated coefficients to transform the control group's outcomes for all time periods. In the periods after treatment, this yields a counterfactual outcome for the treated group had the treatment not occurred—under the three assumptions SUTVA, NPE, and CSB. The average treatment effect on the treated is then estimated as the difference between the observed average outcomes of the treated group and these normalized control outcomes.

⁴The mirror image holds for the selection bias in logs.

We propose a testing procedure to validate the key assumptions of our framework: (i) the presence of selection bias and (ii) the constancy of the selection bias. First, we test whether the normalization coefficients estimated in the first step significantly differ from the theoretical values expected under no selection bias. Second, we assess the stability of these coefficients across multiple pre-treatment periods by conducting joint significance tests to ensure that the normalized pre-treatment trends of the treated and control groups are statistically indistinguishable. As for any *pre-test*, failure in this diagnostic test would call into question the validity of our identifying assumption of constant double selection bias.

We suggest two complementary approaches to ensure that standard errors from the two-step estimation procedure properly capture the first-step uncertainty. First, a non-parametric bootstrap approach resamples the data across both steps, thereby capturing the full estimation uncertainty without relying on strong parametric assumptions. Second, we derive analytical standard errors that explicitly account for the two-step estimation process. Together, these two methods provide robust inference, ensuring that our confidence intervals and hypothesis tests accurately reflect the overall uncertainty in the causal effect estimates.

The versatility of our double selection framework makes it applicable across a wide range of empirical settings. We illustrate its flexibility by revisiting the seminal study by [Card \(1992\)](#) on the labor market consequences of increasing the minimum wage in California in July 1988. In his original study, Card relies on a weighted average of five states that did not experience an increase in the minimum wage as the control group. Using the same data, we first demonstrate that our outcome-on-outcome estimator, while allowing for double selection, supports single selection (as in the traditional DiD framework) and replicates the estimates of the DiD estimator used in the original study. In particular, we find an approximate 5 percent increase in average retail wages in response to the minimum wage increase, consistent with Card’s findings. We then show that an alternative state, Florida, can also be used as a control under double selection, despite not exhibiting constant selection bias in levels (nor logs). Using this alternative control, we estimate a treatment effect of similar magnitude. Further, applying our pre-test to each U.S. state one-by-one as possible controls for California, we find that 23 states satisfy the test. Among these, ten display single selection in levels, one exhibits single selection in logs, and in twelve cases, the pre-test suggests double selection. Thus, the double selection framework effectively doubles the number of possible controls relative to traditional DiD approaches. This expansion is particularly valuable in settings where data availability is limited. For example, consider a scenario where data are available only for the last twelve states. All of these states fail the pre-test in a traditional DiD setting due to absence of parallel trends. Under standard approaches, these states would be discarded as invalid controls, preventing further estimation. In contrast, the more flexible

double selection framework enables the estimation by providing valid counterfactuals based on these same states. In this way, the framework expands the set of admissible controls beyond what traditional DiD methods would allow.

Literature

Our framework relates to the growing literature questioning the robustness of DiD estimates to functional form assumptions. Prior work has emphasized that the choice between levels and logs—while often made ad hoc—has direct implications for the validity of the parallel trends assumption (Finkelstein, 2007; Powell and Leider, 2020; Park and Powell, 2021). Indeed, in his textbook, Cunningham (2021) observes: *"Very little thought if any is given to how exactly we should model some outcome. Just to take one example, should we use the log or the levels themselves? Should we use the quartic root? Should we use rates? These, it turns out, are critically important because for many of them, the parallel trends assumption needed for identification will not be achieved—even though it will be achieved under some other unknown transformation. (...) You must make strong commitments about the functional form itself."*

Our contribution differs from approaches that transform the outcome to achieve identification. For example, Athey and Imbens (2006) and Roth and Sant'Anna (2023) develop models where identification is preserved under specific monotonic transformations. In contrast, we introduce a data-generating process with two types of selection bias, additive and proportional, allowing for identification even when no transformation yields parallel trends.

We also avoid common issues associated with log transformations, such as undefined values at zero, by estimating directly in levels and using an outcome-on-outcome normalization (Meyer, 1995; Chen and Roth, 2024). This distinguishes our method from alternatives such as inverse hyperbolic sine transformations (Bellemare and Wichman, 2020) or PPML (Head and Mayer, 2014), which impose parametric structure on the conditional mean. Instead, our approach builds non-parametrically on group-level outcome dynamics. Revisiting Card (1992), we illustrate how our DS approach indeed expands the cases in which causal inference can be conducted, without relying on a parallel time trend assumption.

The paper is organized as follows. In Section 2, we describe the empirical setting, and introduce the concept of double selection. In Section 3, we prove the identification of treatment effects in the context of double selection. In Section 4, we detail our estimation strategy, testing, and inference. In Section 5, we illustrate our framework. Finally, Section 6 concludes.

2 Empirical Setting: A DGP with Single or Double Selection

Standard DiD models implicitly restrict the form of selection bias to be either additive (in levels) or proportional (in logs), but not both. In this section, we formalize a data-generating process (DGP) that accommodates the simultaneous presence of both forms of selection, what we refer to as *double selection*. This framework nests the traditional single-selection cases as special instances and provides the foundation for the identification and estimation strategy developed in the following sections. We characterize the conditions under which single-selection assumptions emerge and highlight the implicit restrictions they impose. When neither structure fully captures the data, the resulting selection bias is a time-varying linear combination of both types.

2.1 The Data-Generating Process

Let q_{igt} denote the level of some outcome of interest for individual i , belonging to group $g = \{1, \dots, n_g\}$, at time t . In order to allow for a log representation, we restrict $q_{igt} \in \mathbb{R}^{++}$ here, but remove this restriction later. There is a common aggregate outcome component, $\lambda_t \in \mathbb{R} \forall i, g, t$ with $|\lambda_t| < \infty$. This common component shifts outcomes for all groups in the absence of policy treatment, and it can be deterministic or stochastic. At some time t_p a policy is introduced in some groups, which affects the outcome starting in that period in a flexible way. The described DGP can be summarized as

$$q_{igt} = \underbrace{\theta_{0g} + \theta_{1g}\lambda_t + \varepsilon_{igt}}_{=y_{igt}} + \mathbb{1}_{t \geq t_p} \gamma_{gt}, \quad (1)$$

where $\theta_{0g} \in \mathbb{R}$ is a group-specific level shifter and $\theta_{1g} \in \mathbb{R}^{++}$ is a group-specific proportional shifter. Both shifters are constant over time. ε_{igt} is an idiosyncratic shock with $\mathbb{E}_i[\varepsilon_{igt}] = 0$ and constant variance $\text{Var}[\varepsilon_{igt}] = \mathbb{E}_i[\varepsilon_{igt}^2] - \mathbb{E}_i[\varepsilon_{igt}]^2 = \sigma_g^2$ with $0 < \sigma_g^2 < \infty$, where operator $\mathbb{E}_i[\cdot]$ denotes the expectation across individuals i given g and t . Finally, $\mathbb{1}_{t \geq t_p}$ is an indicator function equal to one if $t \geq t_p$, and zero otherwise, and γ_{gt} captures the policy effect for periods $t \geq t_p$ in groups that are treated by a policy change in period t_p . We denote the outcome in the absence of policy by y_{igt} .

2.2 Characterizing the Selection Bias

We next discuss several properties of this DGP, explain how the parameters θ_{0g} and θ_{1g} shape the differences between groups—and in particular, how these differences are captured by generally time-varying selection bias in levels and logs. To lay the groundwork for our analysis of selection bias, we first formalize the relationship between the level and log specifications of the DGP.

Lemma 1 shows how the DGP can be equivalently expressed in multiplicative form under mild conditions, allowing us to characterize group-level differences on both scales and to define bias in levels and logs in a unified framework.

Lemma 1. *Consider the DGP in (1) absent policy. If $y_{igt} = \theta_{0g} + \theta_{1g}\lambda_t + \varepsilon_{igt} \in \mathbb{R}^{++}$, then:*

(a) $\theta_{0g} + \theta_{1g}\lambda_t > 0, \quad \forall(g, t).$

(b) *The individual outcome path absent policy can be equivalently formulated as:*

$$y_{igt} = (\theta_{0g} + \theta_{1g}\lambda_t) u_{igt}, \quad \text{with } u_{igt} = 1 + \frac{\varepsilon_{igt}}{\theta_{0g} + \theta_{1g}\lambda_t} > 0, \quad \text{and } \mathbb{E}_i[u_{igt}] = 1 \quad (2)$$

(c) *For small values of $u_{igt} - 1$, the expectation of $\ln(u_{igt})$ can be approximated as:*

$$\mathbb{E}_i[\ln(u_{igt})] \approx \mathbb{E}_i[u_{igt} - 1]$$

Proof: See Appendix B.1.

Equipped with this connection between an additive error term ε_{igt} and a proportional error term u_{igt} , Definition 1 formalizes the selection bias across groups as the difference between the expected outcome of group g and the expected outcome of a reference group g^* , expressed in levels or logs, and evaluated in the absence of policy.

Definition 1. *The selection bias induced by the DGP in (1), either in levels or logs, is:*

$$\text{Levels: } \mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}] = \theta_{0g} - \theta_{0g^*} + (\theta_{1g} - \theta_{1g^*})\lambda_t \quad (\text{not constant}) \quad (3)$$

$$\begin{aligned} \text{Logs: } & \mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}] \\ &= \mathbb{E}_i[\ln((\theta_{0g} + \theta_{1g}\lambda_t)u_{igt})] - \mathbb{E}_i[\ln((\theta_{0g^*} + \theta_{1g^*}\lambda_t)u_{ig^*t})] \\ &= \ln(\theta_{0g} + \theta_{1g}\lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*}\lambda_t) + \mathbb{E}_i[\ln u_{igt}] - \mathbb{E}_i[\ln u_{ig^*t}] \\ &\approx \ln(\theta_{0g} + \theta_{1g}\lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*}\lambda_t) \quad (\text{not constant}) \end{aligned} \quad (4)$$

where the last line uses the approximation $\mathbb{E}_i[\ln u_{igt}] \approx \mathbb{E}_i[u_{igt} - 1]$ (from Lemma 1(c)), and the value of the expectation $\mathbb{E}_i[u_{igt}] = 1$ (from Lemma 1(b)).

Selection Bias Over Time. Definition 1 directly shows that the selection bias across groups is generally not constant in the presence of both additive and proportional shifters, as long as the

Table 1: Selection Bias in DGP (1), Subcases

$\theta_1 \backslash \theta_0$	$\theta_1 \in \Theta_1 \mathbf{C1}$	$\theta_1 \notin \Theta_1 \mathbf{C1}$
$\theta_0 \in \Theta_0 \mathbf{C0}$	No Selection Corollary 1	Single Selection, Logs Proposition 1(b)
$\theta_0 \notin \Theta_0 \mathbf{C0}$	Single Selection, Levels Proposition 1(a)	Double Selection Proposition 2

Notes: Selection bias arising from DGP (1) where the set $\Theta_0 | \mathbf{C0}$ collects the vector θ_0 for which constraint (C0) holds, and $\Theta_1 | \mathbf{C1}$ collect the vectors θ_1 for which constraint (C1) holds.

common component λ_t varies over time, which we also summarize in Lemma B.1 in Appendix B. Remember that the constancy of selection bias lies at the core of DiD identification. Thus, in order to build intuition for the implications of this non-constancy, we now discuss how the selection terms evolve over time.

To structure this discussion, we consider four different parameter constellations of the DGP in (1), which give rise to distinct patterns of selection bias. Formally, we introduce constraints on vectors $\theta_0 = (\theta_{01}, \dots, \theta_{0n_g}) \in \Theta_0 = \mathbb{R}^{n_g}$ and $\theta_1 = (\theta_{11}, \dots, \theta_{1n_g}) \in \Theta_1 = \mathbb{R}^{n_g, ++}$. We consider two constraints: (C0), which imposes a zero level shifter for all groups, and (C1), which restricts all groups to share the same proportional shifter. Depending on whether constraint (C0), (C1), both, or neither hold, the DGP in Equation (1) gives rise to four distinct patterns of selection bias. In two of those cases there is *single constant selection bias* (single CSB), either in levels or in logs—depending on which of the two constraints holds. In one case there is a *double selection bias*—this arises when none of the constraints hold and both the level and proportional shifters differ between groups. Finally, there is the case of no selection—when both constraints hold. Table 1 summarizes these subcases and refers to Propositions 1 and 2, and Corollary 1, which formally state the implications of the different parameter restrictions for the arising selection bias.

Proposition 1. *Single Constant Selection Bias:* Let (C0) and (C1) denote constraints on the group-specific level shifters and proportional shifters, respectively:

$$\text{Constraint 0 (C0): } \theta_{0g} = \theta_{0g^*} = 0 \ \forall g$$

$$\text{Constraint 1 (C1): } \theta_{1g} = \theta_{1g^*} \ \forall g$$

Let $\Theta_0|\mathbf{C0}$ collect all vectors θ_0 for which constraint (C0) holds, and $\Theta_1|\mathbf{C1}$ collect the vectors θ_1 for which constraint (C1) holds.

- (a) *Levels: If $\theta_0 \notin \Theta_0|\mathbf{C0}$ and $\theta_1 \in \Theta_1|\mathbf{C1}$, then the DGP in (1) exhibits a constant selection bias (CSB) in levels, and not in logs.*
- (b) *Logs: If $\theta_0 \in \Theta_0|\mathbf{C0}$ and $\theta_1 \notin \Theta_1|\mathbf{C1}$, then the DGP in (1) exhibits a constant selection bias (CSB) in logs, and not in levels.*

Proof: see Appendix B.2.

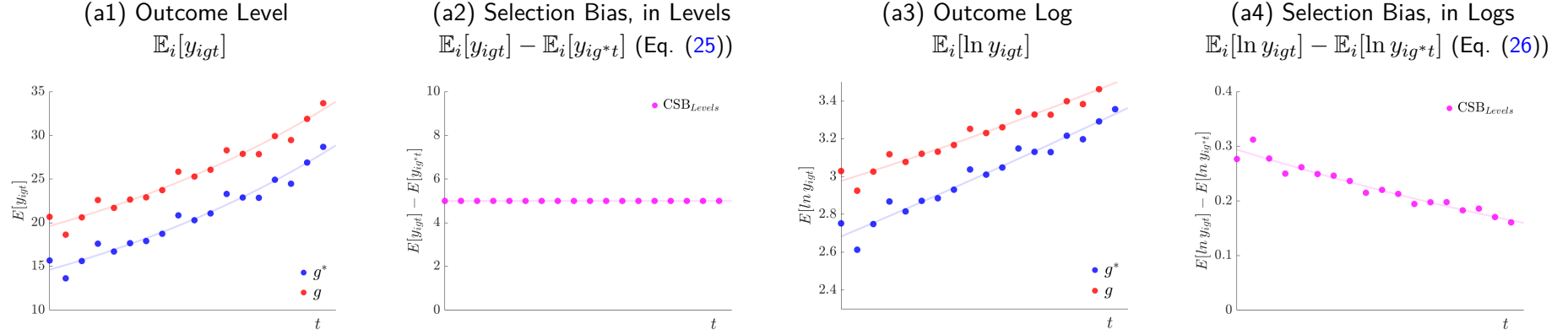
Figure 1 illustrates the implications of Proposition 1 by contrasting two data-generating processes, one with constant selection bias in levels (Panel (a)) and one with constant selection bias in logs (Panel (b)). In both panels, subpanels (1) and (2) display group-level outcomes and selection bias in levels, while subpanels (3) and (4) show the same quantities in logs. In Panel a, the group-specific differences are governed by additive shifts θ_{0g} , with the same proportional component across groups ($\theta_{1g} = \theta_{1g^*}$). As a result, selection bias is constant in levels: the treated and control groups exhibit parallel trajectories in subpanel (1), and the difference in outcomes remains flat in subpanel (2). However, because the proportional gap depends on the scale of the common component λ_t , the log-difference varies over time, as shown in subpanel (4). In contrast, Panel b features a purely proportional shift ($\theta_{0g} = \theta_{0g^*} = 0$, $\theta_{1g} \neq \theta_{1g^*}$), resulting in constant selection bias in logs: the difference in log outcomes remains constant in subpanel (4), while the level difference expands with λ_t , as seen in subpanel (2). These two processes correspond to the single-selection cases characterized in Proposition 1 and highlight that constant selection bias in one scale typically implies non-constant bias in the other.

When both the additive and proportional components are identical across groups, as specified in Corollary 1, the expected outcomes coincide at all points in time, and there is no selection bias. Graphically, this corresponds to treated and control groups tracing out the same path in the absence of treatment. While this case rarely holds in practice, it provides a natural benchmark for comparing the single- and double-selection scenarios.

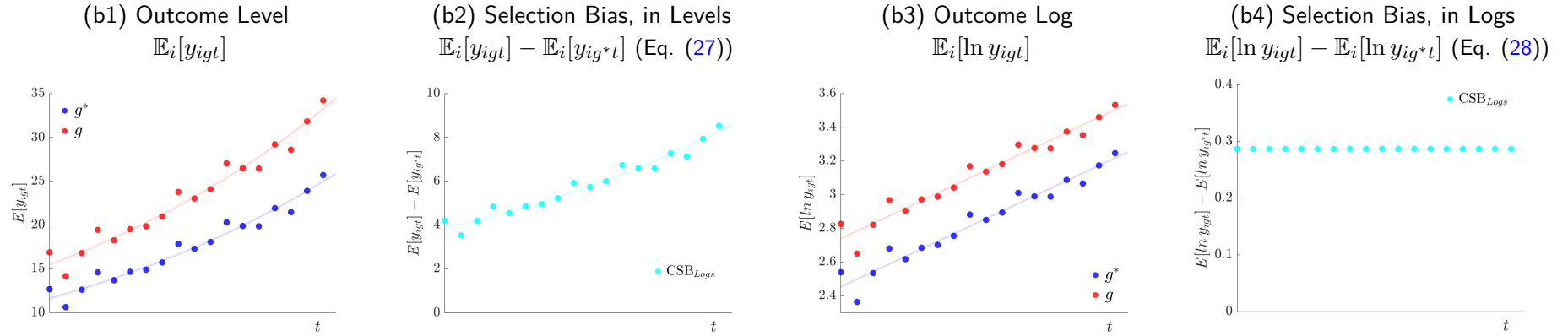
Corollary 1. *If $\theta_0 \in \Theta_0|\mathbf{C0}$ and $\theta_1 \in \Theta_1|\mathbf{C1}$, then there is neither selection in levels nor in logs.*

Figure 1: Single Selection

(a) Single CSB in Levels: Constraint **(C1)** ($\theta_{1g} = \theta_{1g^*} \forall g$)



(b) Single CSB in Logs: Constraint **(C2)** ($\theta_{0g} = \theta_{0g^*} = 0 \forall r$)



Notes: $\lambda_t = \lambda_0(1 + l_\lambda)^t + \eta_t$ with $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$. Lines plot the deterministic component of the DGP; scattered dots plot the DGP including the stochastic component η_t . The axes are adjusted to ease the visualization of either CTT in levels or logs. (Parameters: $\lambda_0 = 3$, $l_\lambda = 0.001$, $\sigma_\eta^2 = 0.25$; panel (a): $\theta_{0g^*} = 3$, $\theta_{1g^*} = \theta_{1g} = 1.1$, $\theta_{0g} = 8$; panel (b): $\theta_{0g^*} = \theta_{0g} = 0$, $\theta_{1g^*} = 1.1$, $\theta_{1g} = 1.465$.)

The taxonomy defined by Proposition 1 and Corollary 1 delineates the special cases in which selection bias is constant in either levels or logs, or absent altogether. When neither constraint is imposed, group differences may reflect both additive and proportional components. In this more general case, which we refer to as double selection, the selection bias becomes a time-varying linear combination of the two single-selection components. Proposition 2 formalizes this result.

Proposition 2. Double Selection: *If $\theta_0 \notin \Theta_0|C0$ and $\theta_1 \notin \Theta_1|C1$, then*

- (a) *The selection bias in levels in (3) is a linear combination of the selection bias that arises under constraint (C0) and the selection bias that arises under constraint (C1). The weights on the two terms vary with the common component λ_t .*
- (b) *The selection bias in logs in (4) is a linear combination of the selection bias that arises under constraint (C0) and the selection bias that arises under constraint (C1). The weights on the two terms vary with the common component λ_t .*

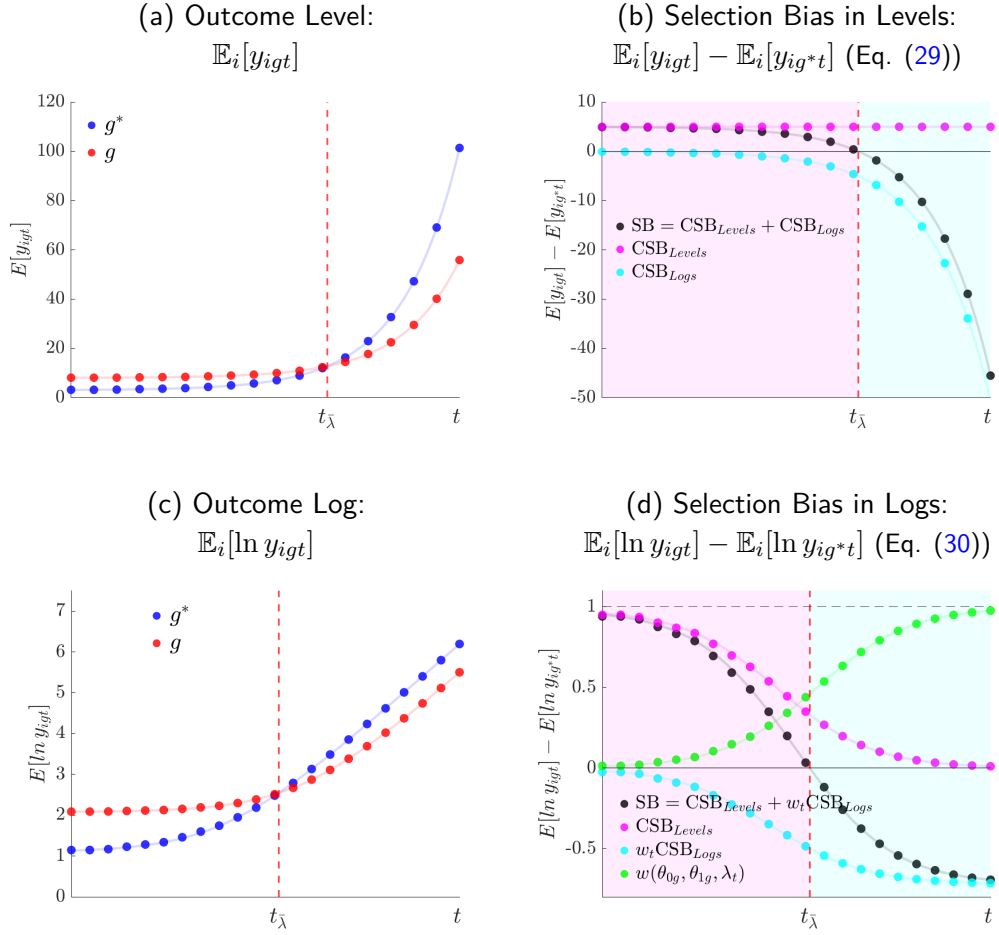
Proof: see Appendix B.3.

Figures 2 and 3 illustrate the essence of Proposition 2: when both additive and proportional group-specific components are active, selection bias is generally not constant in either levels or logs. Figure 2 presents a simulated example with a deterministic common component λ_t , where the treated group has a larger level shifter but a smaller proportional shifter than the control. This configuration produces time-varying and non-monotonic selection bias in both levels (panels (a) and (b)) and logs (panels (c) and (d)), with clear sign reversals over time as the proportional component begins to dominate. Figure 3 confirms that these patterns persist under alternative specifications of λ_t : one deterministic, where both types of selection go in the same direction, and one stochastic. In both cases, selection bias varies systematically with the common component, rather than following a constant trend.

This reflects a more general feature of double selection: The size of the group differences in the level shifters and the proportional shifters, as well as the magnitude of λ_t , determine whether selection in levels or selection in logs dominates. The larger the common component, the more relevant the group differences in proportional effects. This induces the potential for non-monotonic behavior of the selection bias over time, and for sign reversals in selection bias, even in the absence of policy treatment. We formalize this intuition in Appendix B.4, where we derive conditions under which one source of selection dominates.⁵

⁵Lemma B.2 characterizes the slope of the selection bias with respect to λ_t ; Lemma B.3 provides limiting values as $\lambda_t \rightarrow 0$ or ∞ ; and Propositions B.1 and B.2 formalize the conditions under which one component dominates.

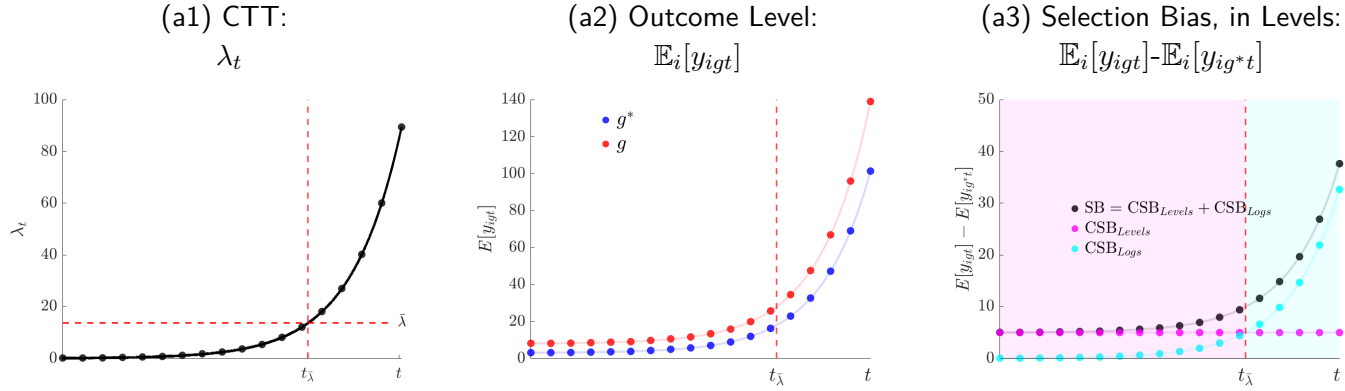
Figure 2: Double Selection



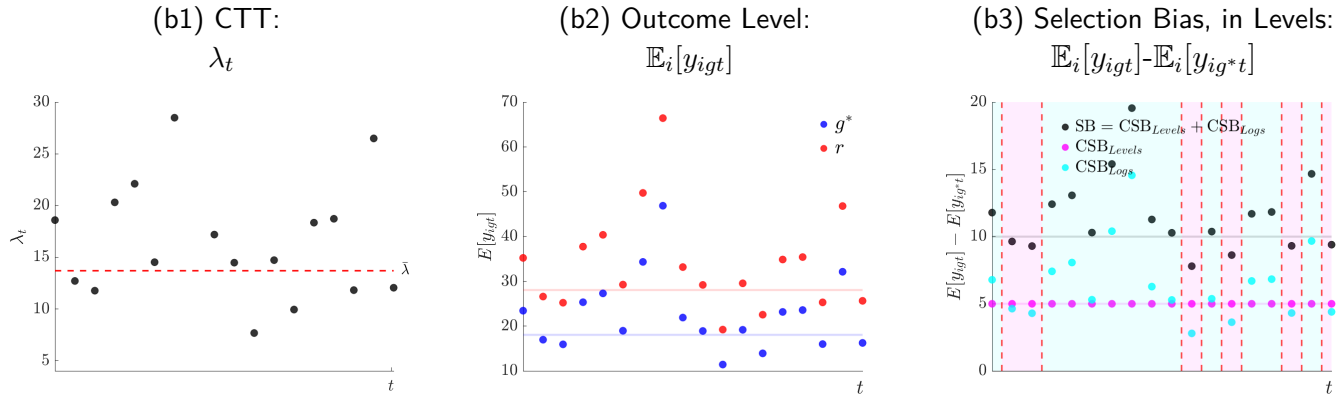
Notes: $\lambda_t = \lambda_0(1 + l_\lambda)^t + \eta_t$, with $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$. The figure illustrates double selection in opposite directions, with $\theta_{0g} > \theta_{0g^*}$ and $\theta_{1g^*} > \theta_{1g} > 0$. Lines plot the deterministic component of the DGP; scattered dots plot the DGP including the stochastic component η_t . The axes are adjusted to ease the visualization. (Parameters: $\lambda_0 = 3$, $l_\lambda = 0.001$, $\sigma_\eta^2 = 0.25$; panel (a): $\theta_{0g^*} = 3$, $\theta_{1g^*} = 1.1$, $\theta_{0g} = 8$, $\theta_{1g} = 1.465$, $\bar{\lambda} = 13.69$, $t_{\bar{\lambda}} = 4922.3$; panel (b): $\theta_{1g} = 0.535$, $\bar{\lambda} = 8.84$, $t_{\bar{\lambda}} = 4485.2$.)

Figure 3: Double Selection: Illustrating Alternative Common Time Trends λ_t

(a) A Deterministic CTT: $\lambda_t = \lambda_0(1 + l_\lambda)^t + \eta_t$ and $\eta_t \sim iid(0, \sigma_\eta^2)$



(b) A Stochastic CTT: $\lambda_t = \bar{\lambda}e^{\eta_t}$ and $\eta_t \sim iid(0, \sigma_\eta^2)$



Notes: $\theta_{0g} > \theta_{0g^*}$, $\theta_{1g} > \theta_{1g^*} > 0$. Lines plot the deterministic component of the DGP; scattered dots plot the DGP including the stochastic component η_t . The axes are adjusted to ease the visualization. (Parameters: $\theta_{0g^*} = 3$, $\theta_{1g^*} = 1.1$, $\theta_{0g} = 8$, $\theta_{1g} = 1.465$, $\bar{\lambda} = 13.69$; panel (a): $\lambda_0 = 3$, $l_\lambda = 0.001$, $\sigma_\eta^2 = 0.25$; panel (b): $\sigma_\eta^2 = 1.96$.)

2.3 Outcome-on-Outcome Transformation

Although the relative importance of level and log selection varies with the common component, and thus varies over time, the structure of the DGP imposes a stable relationship between any two groups. The expected outcome of one group is a linear transformation of the outcome of another. This insight motivates a shift in perspective: rather than comparing the outcomes of treated and control groups in levels or logs in the cross-section, we examine how the outcome of one group varies with the outcome of the other. Proposition 3 formalizes the result and shows that the slope and intercept of the linear mapping reflect the differences between groups in the level and proportional shifters.

Proposition 3. Linear Relationship in Outcome-on-Outcome Space: *For any two groups, g and g^* , that follow DGP (1), the expected outcome of one group can be expressed as a linear transformation of the expected outcome of the other group: $\mathbb{E}_i[y_{igt}] = \omega_0 + \omega_1 \mathbb{E}_i[y_{ig^*t}]$. The coefficients (ω_0, ω_1) are pinned down by differences in the selection parameters. This linear relationship holds regardless of the evolution over time of the common component λ_t .*

Proof. Taking the expectation over individuals in (1) for groups g and g^* , we obtain

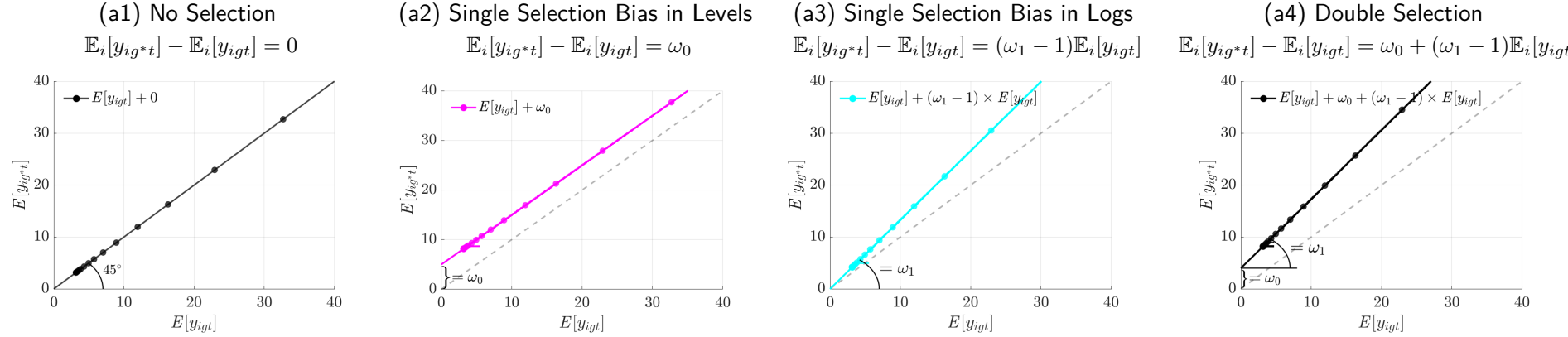
$$\mathbb{E}_i[y_{igt}] = \theta_{0g} + \theta_{1g}\lambda_t \quad \text{and} \quad \mathbb{E}_i[y_{ig^*t}] = \theta_{0g^*} + \theta_{1g^*}\lambda_t.$$

Combining the two equations yields $\mathbb{E}_i[y_{ig^*t}] = \theta_{0g^*} - \frac{\theta_{1g^*}}{\theta_{1g}}\theta_{0g} + \frac{\theta_{1g^*}}{\theta_{1g}}\mathbb{E}_i[y_{igt}] = \omega_0 + \omega_1 \mathbb{E}_i[y_{igt}]$. $\square$

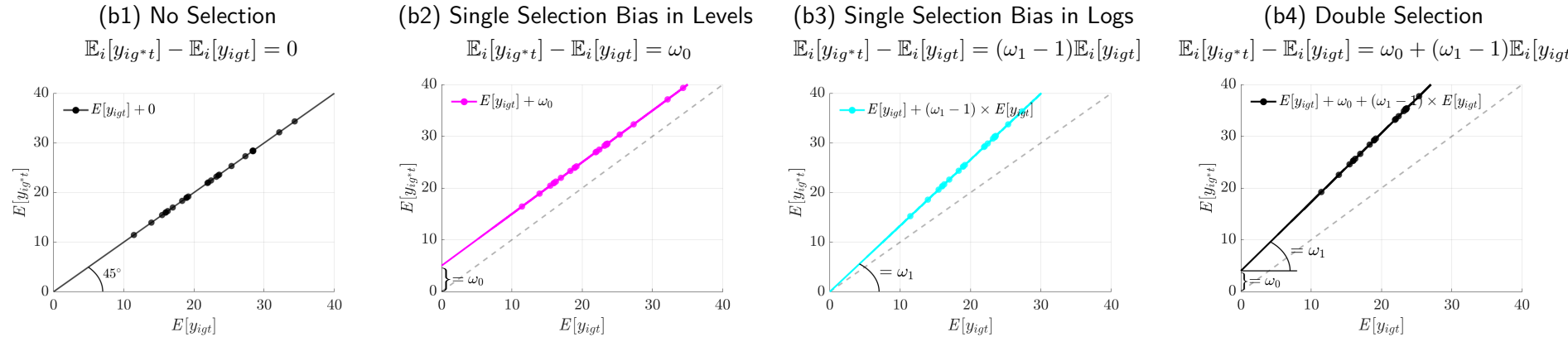
Figure 4 illustrates how the outcome-on-outcome mapping reflects the presence and type of selection across groups for the four cases outlined in Table 1. The two panels of the figure reflect two different examples for the common component: in panel (a) λ_t features a strong deterministic component, while in panel (b) λ_t is purely stochastic. In the second case, the randomness of the common component gives rise to selection bias which evolves non-monotonically over time, as we discussed before. However, in all four different cases of selection bias, the outcome-on-outcome relationship remains exactly linear, as implied by Proposition 3. In the absence of selection ($\theta_{0g} = \theta_{0g^*} = 0$, $\theta_{1g} = \theta_{1g^*}$) expected outcomes for groups g and g^* lie on a 45° line, as shown in panels (a1) and (b1). Under single selection in levels ($\theta_{0g} \neq \theta_{0g^*}$, $\theta_{1g} = \theta_{1g^*}$), the intercept of the linear mapping is non-zero, which shifts the line vertically while preserving the slope. This is depicted in panels (a2) and (b2). Under single selection in logs ($\theta_{0g} = \theta_{0g^*} = 0$, $\theta_{1g} \neq \theta_{1g^*}$), the slope deviates from one, while the intercept remains anchored at zero, shown in Panels (a3) and (b3). Finally, under double selection ($\theta_{0g} \neq \theta_{0g^*}$, $\theta_{1g} \neq \theta_{1g^*}$), as shown in panels (a4) and (b4), the mapping exhibits both a shifted intercept and a non-unit slope.

Figure 4: Visualizing Double Selection in the Outcome-on-Outcome Space: $\mathbb{E}_i[y_{g^*t}]$ against $\mathbb{E}_i[y_{igt}]$

(a) A Deterministic CTT: $\lambda_t = \lambda_0(1 + l_\lambda)^t + \eta_t$ and $\eta_t \sim iid(0, \sigma_\eta^2)$



(b) A Stochastic CTT: $\lambda_t = \bar{\lambda}e^{\eta_t}$ and $\eta_t \sim iid(0, \sigma_\eta^2)$



Notes: $\theta_{g0} > \theta_{g^*0}$, $\theta_{g1} > \theta_{g^*1} > 0$. Lines plot the deterministic component of the DGP; scattered dots plot the DGP including the stochastic component η_t . (Parameters: $\theta_{g^*0} = 3$, $\theta_{g^*1} = 1.1$, $\theta_{g0} = 8$, $\theta_{g1} = 1.465$, $\bar{\lambda} = 13.69$; panel (a): $\lambda_0 = 3$, $l_\lambda = 0.001$, $\sigma_\eta^2 = 0.25$; panel (b): $\sigma_\eta^2 = 1.96$.)

3 Identification Under Double Selection

Again, we consider the outcome q_{igt} of an individual i belonging to group g at time t . Let $\mathbf{P}_{gt} \in \{0, 1\}$ denote the treatment status of a group g in period t , i.e., $\mathbf{P}_{gt}$ takes on value 1 if in period t group g is subject to the policy of interest, and value 0 otherwise. We consider a setting where no-one is subject to the policy initially, before, starting in some period t_p , a group is subject to a policy change, and a control group is not. This treatment status is captured by a binary variable $\mathbf{P}_g$:

$$\mathbf{P}_g = \begin{cases} 1 & \text{if } \mathbf{P}_{gt} = 0 \forall t < t_p \wedge \mathbf{P}_{gt} = 1 \forall t \geq t_p \\ 0 & \text{if } \mathbf{P}_{gt} = 0 \forall t. \end{cases} \quad (5)$$

We refer to the two groups by $\mathcal{T}$ ($\mathbf{P}_{\mathcal{T}} = 1$) and $\mathcal{C}$ ($\mathbf{P}_{\mathcal{C}} = 0$).

Making use of the potential outcome notation (cf. [Imbens and Rubin, 2015](#)), we assume that for each possible treatment status there is a potential outcome, which we denote by $q_{igt}(1)$ and $q_{igt}(0)$, respectively. Our object of interest is the true policy at some time $t \geq t_p$, $q_{igt}(1) - q_{igt}(0)$. The identification challenge is that we observe an individual only as either treated or untreated. In Assumption 1, we establish a set of identification assumptions under which the policy effect is identified from observable data for a set of individuals, some of which are treated and some of which are not, which we state in Proposition 4. Precisely, we are interested in the identification of the Average Treatment Effect on the Treated (ATT) in period t :

$$ATT(t) = \mathbb{E}_i[q_{igt}(1) - q_{igt}(0) | g = \mathcal{T}] \text{ for } t \geq t_p, \quad (6)$$

where the expectation operator is with respect to individuals i , taken here over all individuals within the treated group $\mathcal{T}$.

Assumption 1. *The DGP of q_{igt} satisfies the following three assumptions.*

(IA1) *Stable Unit Treatment Value Assumption (SUTVA): We assume that the treatment status $\mathbf{P}_g$ uniquely pins down which of the two possible potential outcomes is observed for any t , and that the treatment status of any other unit is irrelevant,*

$$q_{igt} = q_{igt}(\mathbf{P}_g). \quad (7)$$

(IA2) *No Pre-Treatment Effects (NPE)*: we assume that for any t before the policy date t_p , the two potential outcomes are the same,

$$\mathbb{E}_i[q_{igt}(0)|\mathcal{T}] = \mathbb{E}_i[q_{igt}(1)|\mathcal{T}] \text{ for } t < t_p. \quad (8)$$

(IA3) *Constant (double) Selection Bias (CSB)*: we assume a constant ratio (between groups) of changes of the potential no-treatment outcome,

$$\frac{\mathbb{E}_i[q_{igt}(0)|\mathcal{T}] - \mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{T}]}{\mathbb{E}_i[q_{igt}(0)|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{C}]} = \phi \text{ for } t \geq t_p \quad (9)$$

where $\phi = \begin{cases} 1, & \text{if } \forall \tau \in \mathbb{N}^+ : \frac{\mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{T}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}(0)|\mathcal{T}]}{\mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}(0)|\mathcal{C}]} = 0 \\ \frac{\mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{T}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}(0)|\mathcal{T}]}{\mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}(0)|\mathcal{C}]} \text{ for some } \tau \in \mathbb{N}^+, & \text{else.} \end{cases}$

Note that if $\phi = 1$, which is the case in a setting with either of the two groups displaying a flat outcome time path before policy, (9) simplifies to the standard assumption of a Common Time Trend in levels.

Proposition 4. Identification of Policy Effect: If a DGP satisfies assumptions (IA1)–(IA3), then the policy effect in (6) (ATT) is identified from observable outcomes of the treated and untreated groups.

Proof. By assumption (IA1), SUTVA, the observed outcome for group $\mathcal{T}$ is equal to the potential outcome under treatment. In order to identify the policy effect in (6) from observables, we need to establish that under assumptions (IA1)–(IA3), the expected no-treatment outcome $\mathbb{E}_i[q_{igt}(0)]$ can be replaced by observables.

Starting from (IA3), CSB, gives an expression for $\mathbb{E}_i[q_{igt}(0)]$ for $t \geq t_p$:

$$\mathbb{E}_i[q_{igt}(0)|\mathcal{T}] = \mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{T}] + \phi \left(\mathbb{E}_i[q_{igt}(0)|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{C}] \right) \text{ for } t \geq t_p.$$

Before turning to the expression for ϕ , note that by (IA2), NPE, the potential no-treatment outcome of the treated group $\mathcal{T}$ is equal to the treatment outcome for $t < t_p$, which gives

$$\mathbb{E}_i[q_{igt}(0)|\mathcal{T}] = \mathbb{E}_i[q_{ig,t_p-1}(1)|\mathcal{T}] + \phi \left(\mathbb{E}_i[q_{igt}(0)|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1}(0)|\mathcal{C}] \right) \text{ for } t \geq t_p.$$

Then, by (IA1), SUTVA, we note that the observed outcomes for the groups $\mathcal{T}$ and $\mathcal{C}$ correspond to the potential outcomes under treatment and no treatment, respectively. This implies that

$$\mathbb{E}_i[q_{igt}(0)|\mathcal{T}] = \mathbb{E}_i[q_{ig,t_p-1}|\mathcal{T}] + \phi \left(\mathbb{E}_i[q_{igt}|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1}|\mathcal{C}] \right) \text{ for } t \geq t_p. \quad (10)$$

We can then use the same two assumptions (IA2) and (IA1) in the expression for ϕ , which in the above equation is then given by

$$\text{where } \phi = \begin{cases} 1, & \text{if } \forall \tau \in \mathbb{N}^+ : \frac{\mathbb{E}_i[q_{ig,t_p-1}|\mathcal{T}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}|\mathcal{T}]}{\mathbb{E}_i[q_{ig,t_p-1}|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}|\mathcal{C}]} = 0 \\ \frac{\mathbb{E}_i[q_{ig,t_p-1}|\mathcal{T}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}|\mathcal{T}]}{\mathbb{E}_i[q_{ig,t_p-1}|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1-\tau}|\mathcal{C}]} & \text{for some } \tau \in \mathbb{N}^+, \text{ else.} \end{cases}$$

Thus, under (IA1)–(IA3), observable outcomes of groups $\mathcal{T}$ and $\mathcal{C}$ identify the expected outcomes under both, treatment and no treatment, for group $\mathcal{T}$, which identifies the ATT in (6). $\square$

For illustration of identification, consider the benchmark DGP of equation (1), which satisfies Assumption 1, and which we repeat here for convenience:

$$q_{igt} = \theta_{0g} + \theta_{1g}\lambda_t + \mathbf{P}_{gt}\gamma_{gt} + \varepsilon_{it}$$

for individuals in group g . With this DGP, the potential outcomes under treatment and no treatment are given by

$$q_{igt}(0) = \theta_{0g} + \theta_{1g}\lambda_t + \varepsilon_{it} \quad \text{and} \quad q_{igt}(1) = \theta_{0g} + \theta_{1g}\lambda_t + \gamma_{gt} + \varepsilon_{it},$$

and thus the *ATT* in period t is given by

$$ATT(t) = \mathbb{E}_i[q_{igt}(1) - q_{igt}(0)|g = \mathcal{T}] = \gamma_{\mathcal{T}t} \text{ for } t \geq t_p. \quad (11)$$

Use the DGP in (10) (i.e., in the expression that replaces the unobserved no-treatment outcome for the treated group with observables) to obtain

$$\begin{aligned} \mathbb{E}_i[q_{igt}(0)|\mathcal{T}] &= \mathbb{E}_i[q_{ig,t_p-1}|\mathcal{T}] + \phi \left(\mathbb{E}_i[q_{igt}|\mathcal{C}] - \mathbb{E}_i[q_{ig,t_p-1}|\mathcal{C}] \right) \\ &= \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_{t_p-1} + \phi \left(\theta_{0\mathcal{C}} + \theta_{1\mathcal{C}}\lambda_t - \theta_{0\mathcal{C}} - \theta_{1\mathcal{C}}\lambda_{t_p-1} \right) \\ &= \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_{t_p-1} + \phi\theta_{1\mathcal{C}} \left(\lambda_t - \lambda_{t_p-1} \right) \text{ for } t \geq t_p, \end{aligned} \quad (12)$$

where for $\theta_{1\mathcal{T}} \neq 0$ and $\theta_{1\mathcal{C}} \neq 0$, ϕ is given by

$$\frac{\theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_{t_p-1} - \theta_{0\mathcal{T}} - \theta_{1\mathcal{T}}\lambda_{t_p-1-\tau}}{\theta_{0\mathcal{C}} + \theta_{1\mathcal{C}}\lambda_{t_p-1} - \theta_{0\mathcal{C}} - \theta_{1\mathcal{C}}\lambda_{t_p-1-\tau}} = \frac{\theta_{1\mathcal{T}}\lambda_{t_p-1} - \theta_{1\mathcal{T}}\lambda_{t_p-1-\tau}}{\theta_{1\mathcal{C}}\lambda_{t_p-1} - \theta_{1\mathcal{C}}\lambda_{t_p-1-\tau}} = \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}}, \quad (13)$$

which back in (12) implies

$$\mathbb{E}_i[q_{igt}(0)|\mathcal{T}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_{t_p-1} + \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}}\theta_{1\mathcal{C}}(\lambda_t - \lambda_{t_p-1}) \text{ for } t \geq t_p. \quad (14)$$

Subtracting this from the observed treatment outcome of group $\mathcal{T}$, we obtain that the ATT for some $t \geq t_p$ is identified from observables as:

$$\begin{aligned} ATT(t) &= \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_t + \gamma_{\mathcal{T}t} - \theta_{0\mathcal{T}} - \theta_{1\mathcal{T}}\lambda_{t_p-1} - \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}}\theta_{1\mathcal{C}}(\lambda_t - \lambda_{t_p-1}) \\ &= \gamma_{\mathcal{T}t} + \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}(\lambda_t - \lambda_{t_p-1}) - \theta_{0\mathcal{T}} - \theta_{1\mathcal{T}}(\lambda_t - \lambda_{t_p-1}) \\ &= \gamma_{\mathcal{T}t} \end{aligned}$$

Note that the above described procedure establishes identification for the different cases discussed in Section 2, and thus covers single selection bias, in levels or logs, and double selection bias, which combines the two. While we express the period- t average treatment effect, $ATT(t)$, as additive in (1), we can translate it into a proportional difference. This is similar to the usual ways of expressing an average treatment effect in levels as percentage of the mean value of the control group (e.g., [Chen and Roth, 2024](#)).

4 Estimation

In this section, we introduce an estimator for the average treatment effect on the treated $ATT(t) = \gamma_t$ that builds on the identifying assumptions outlined in the preceding section and proceeds in two steps. In the first step, we use pre-treatment outcome time paths of both groups to estimate two normalization parameters. These capture two selection biases, and are estimated as coefficients in a regression of the average outcome of the control group on the average outcome of the treatment group. In the second step, we apply these normalization parameters to post-treatment control group outcomes. This yields the estimate of the counterfactual no-treatment outcome of the treated. Using this normalized outcome of the control group, we can estimate the policy effect within a standard Difference-in-Differences regression framework. Further, within this second-step regression framework, we can also resort to otherwise standard

Table 2: Example with Two Groups and Three Periods

Panel A: Outcomes by group over time			
Period Group	$t = 1$ (<i>pre-policy</i>)	$t = 2$ (<i>pre-policy</i>)	$t = 3$ (<i>post-policy</i>)
$\mathcal{C}$	$\mathbb{E}_i[q_{i\mathcal{C}1}] = \theta_{0\mathcal{C}} + \theta_{1\mathcal{C}}\lambda_1$	$\mathbb{E}_i[q_{i\mathcal{C}2}] = \theta_{0\mathcal{C}} + \theta_{1\mathcal{C}}\lambda_2$	$\mathbb{E}_i[q_{i\mathcal{C}3}] = \theta_{0\mathcal{C}} + \theta_{1\mathcal{C}}\lambda_3$
$\mathcal{T}$	$\mathbb{E}_i[q_{i\mathcal{T}1}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_1$	$\mathbb{E}_i[q_{i\mathcal{T}2}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_2$	$\mathbb{E}_i[q_{i\mathcal{T}3}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_3 + \gamma$

Panel B: Transformed outcomes for the control group			
Period Group	$t = 1$ (<i>pre-policy</i>)	$t = 2$ (<i>pre-policy</i>)	$t = 3$ (<i>post-policy</i>)
$\mathcal{C}$	$\mathbb{E}_i[\tilde{q}_{i\mathcal{C}1}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_1$	$\mathbb{E}_i[\tilde{q}_{i\mathcal{C}2}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_2$	$\mathbb{E}_i[\tilde{q}_{i\mathcal{C}3}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_3$
$\mathcal{T}$	$\mathbb{E}_i[q_{i\mathcal{T}1}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_1$	$\mathbb{E}_i[q_{i\mathcal{T}2}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_2$	$\mathbb{E}_i[q_{i\mathcal{T}3}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}}\lambda_3 + \gamma$

Notes: the table uses the same notation as in the DGP described in (1). λ_t represents time effects, θ_{0g} and θ_{1g} are level and proportional selection parameters, respectively. There are three periods and only $t = 3$ is post-policy. γ is the treatment effect of interest.

approaches to pre-testing. Precisely, we can test the null hypothesis of constant double selection in the pre-treatment observables.

A Simple Example. We start with a simple linear example that satisfies (IA1) through (IA3) to illustrate our estimation approach. In Table 2, we consider outcomes for two groups over three time periods: two pre-policy periods and one post-policy period. The treated and control groups share the same aggregate time shocks, λ_t , which for each group g translates linearly into the group-level average outcome. The group-specific constant θ_{0g} and slope coefficient θ_{1g} induce the two types of constant selection bias in the form of group-specific level and proportional differences. In period $t = 3$, the outcome of treatment group $\mathcal{T}$ is subject to the policy, where the policy effect is γ .

Now recall from Proposition 3 and the corresponding visualization in Figure 4 that, given the common time effects λ_t , we can express $\mathbb{E}_i[q_{igt}|\mathcal{T}]$ as a linear transformation of $\mathbb{E}_i[q_{igt}|\mathcal{C}]$ for

every pre-treatment period t :

$$\mathbb{E}_i[q_{igt}|\mathcal{T}] = \underbrace{\left(\theta_{0\mathcal{T}} - \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}} \theta_{0\mathcal{C}}\right)}_{\omega_{\mathcal{C}0}} + \underbrace{\frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}}}_{\omega_{\mathcal{C}1}} \mathbb{E}_i[q_{igt}|\mathcal{C}] \text{ for } t = 1, 2. \quad (15)$$

This gives rise to the first step of the estimator: we can obtain the two coefficients $\omega_{\mathcal{C}0}$ and $\omega_{\mathcal{C}1}$ from a linear regression of $\mathbb{E}_i[q_{igt}|\mathcal{T}]$ on $\mathbb{E}_i[q_{igt}|\mathcal{C}]$, which can be estimated using ordinary least squares (OLS) as the solution to the minimization problem:

$$\{\omega_{\mathcal{C}0}, \omega_{\mathcal{C}1}\} = \arg \min ||\mathbb{E}_i[q_{igt}|\mathcal{T}] - \underbrace{(\omega_{\mathcal{C}0} + \omega_{\mathcal{C}1} \mathbb{E}_i[q_{igt}|\mathcal{C}])}_{=\tilde{q}_{\mathcal{C}t}}|| \text{ for } t < t_p. \quad (16)$$

By using *only* pre-policy data, we ensure that $\hat{\omega}$ captures only the selection parameters. Note that the parameters $\omega_{\mathcal{C}0}$ and $\omega_{\mathcal{C}1}$ are identified through variations of average outcomes between the treatment and control groups over pre-policy periods. This is what makes two pre-policy periods the minimum requirement for being able to pin down the selection parameters. Intuitively, $\omega_{\mathcal{C}0}$ captures baseline differences in levels, while $\omega_{\mathcal{C}1}$ accounts for proportional differences.

For the second step of the estimator, we then use these obtained coefficients to construct a transformed mean outcome of the control group for all periods:

$$\tilde{q}_{\mathcal{C}t} = \omega_{\mathcal{C}0} + \omega_{\mathcal{C}1} \mathbb{E}_i[q_{igt}|\mathcal{C}]. \quad (17)$$

Using the coefficients that solve the minimization problem, $\omega_{\mathcal{C}0} = \left(\theta_{0\mathcal{T}} - \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}} \theta_{0\mathcal{C}}\right)$ and $\omega_{\mathcal{C}1} = \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}}$, we obtain $\tilde{q}_{\mathcal{C}t} = \left(\theta_{0\mathcal{T}} - \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}} \theta_{0\mathcal{C}}\right) + \frac{\theta_{1\mathcal{T}}}{\theta_{1\mathcal{C}}} \mathbb{E}_i[q_{igt}|\mathcal{C}] = \theta_{0\mathcal{T}} + \theta_{1\mathcal{T}} \lambda_t$. In this example, the transformed outcomes for the control group satisfy $\mathbb{E}_i[\tilde{q}_{it}|\mathcal{C}] = \mathbb{E}_i[q_{igt}|\mathcal{T}]$ in every pre-policy period ($t = 1, 2$) by construction. Likewise, the transformed outcome corresponds to the counterfactual outcome of the treatment group under no treatment in period $t = 3$, as shown in the lower panel of Table 2. This illustrates that, after applying the normalization coefficients, which are obtained using only pre-policy time periods, to the post-policy time period ($t = 3$), the difference between the outcome of the treated group and the normalized outcome of the control group pins down the treatment effect γ . This holds because the normalization eliminates both level and proportional selection effects, ensuring that the outcomes for the transformed control group provide a valid counterfactual for the treated group in the absence of treatment—given constant double selection.

4.1 The Estimator

As discussed above, the estimator proceeds in two steps. The first step estimates the normalization parameters, ω_0 and ω_1 , which align the control group's outcomes with those of the treated group in the pre-policy period. The second step applies the normalization to pre- and post-policy data, and estimates the average treatment effect on the treated group using a standard Dynamic DID specification on the transformed outcomes.

Step 1: Normalization of Outcomes

For a sample of n individuals observed over T periods, where T^{pre} periods are pre-policy, we regress average outcomes of the treatment group on those of the control. Let $\mathbf{q}_g$ denote the vector of length T^{pre} , which collects the group-specific mean outcomes, $\mathbb{E}_i[q_{igt}|g]$, for every $t \in \{1, \dots, t_p - 1\}$. The first-step estimation is summarized in the following linear specification:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\omega} + \boldsymbol{\epsilon}, \quad \text{where } \mathbf{Y} = \mathbf{q}_T, \quad \mathbf{X} = [\mathbf{1}, \mathbf{q}_C], \quad \boldsymbol{\omega} = [\omega_{C0}, \omega_{C1}]'. \quad (18)$$

Here, $\mathbf{X}$ includes a constant and the mean outcomes for the control group, while $\mathbf{Y}$ contains the mean outcomes for the treated group in each pre-policy period, and $\boldsymbol{\epsilon}$ is a vector of error terms. The normalization coefficients $\boldsymbol{\omega}$ can then be estimated by OLS as $\hat{\boldsymbol{\omega}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$. This normalization ensures that the transformed control outcomes align with the treated group outcomes before the policy, allowing valid comparisons in the post-policy period.

Step 2: Estimation of Policy Effect

We now use the estimated parameters $\hat{\omega}_{C0}$ and $\hat{\omega}_{C1}$ to construct a normalized outcome of group C for all $t \in 1, \dots, T$, i.e., including the post-policy periods.

$$\hat{q}_{Ct} = \hat{\omega}_{C0} + \hat{\omega}_{C1}\mathbb{E}_i[q_{igt}|\mathcal{C}]. \quad (19)$$

This normalized outcome delivers the counterfactual for the treated group in the periods after policy, so that we obtain the estimate of the policy effect for all $t \geq t_p$ as:

$$\hat{\gamma}_{Tt} = \mathbb{E}_i[q_{igt}|\mathcal{T}] - \hat{q}_{Ct}. \quad (20)$$

Given the normalized outcomes, this second-step estimator can be captured within a standard event-study regression framework. In order to specify the regression at the individual level, we

apply the normalization to individual-level outcomes within the control group.⁶ In particular, we define the adjusted individual-level outcome for individual i by

$$\tilde{q}_{igt} \equiv \begin{cases} q_{igt} & \text{if } g = \mathcal{T} \\ \hat{\omega}_{C0} + \hat{\omega}_{C1}q_{igt} & \text{if } g = \mathcal{C} \end{cases}, \quad (21)$$

and use this in the following regression:

$$\tilde{q}_{igt} = \phi_{0g} + \phi_{1t} + \sum_{t \neq (t_p-1)} \gamma_{\mathcal{T}t} D_{gt} + \epsilon_{igt}. \quad (22)$$

The term ϕ_{0g} is a shorthand notation that captures unobserved heterogeneity at the group level, which absorbs both the global intercept and group-level deviations from it. Similarly, ϕ_{1t} represents time fixed effects, estimated relative to a base period. Finally, $D_{gt} = \mathbb{I}[t, g = \mathcal{T}]$ denotes a treatment dummy interacted with time fixed effects. Due to the normalization procedure, which addresses the selection in the pre-policy period, the time and group effects should be identical in that period for the treated group and the normalized control group. Consequently, the estimates for $\gamma_{\mathcal{T}t}$ are expected to be statistically equal to zero for all $t < t_p - 1$. For $t \geq t_p$, the estimates for $\gamma_{\mathcal{T}t}$ capture the average treatment effect on the treated.

4.2 Testing for Selection

We suggest two complementary tests to assess the validity of our two-step estimator. Both tests build around the estimates of $\hat{\omega} = \{\hat{\omega}_{C0}, \hat{\omega}_{C1}\}$, the two first-step coefficients which capture the (double) selection bias. In the description of these tests, we take the covariance structure of the estimates as given. We will turn to this in the next Section.

The first test examines if these parameters differ significantly from what would be expected under no selection bias. The coefficients under no selection bias are $\hat{\omega}_{C1} = 1$ and $\hat{\omega}_{C0} = 0$. A joint F-test that rejects the null hypothesis $\{\omega_{C0}, \omega_{C1}\} = \{0, 1\}$ thus implies the presence of selection bias. Importantly, as we outlined in Table 1, this can be either level selection, or proportional selection, or both. While four cases are theoretically possible, we cannot distinguish all of them. Not rejecting the null of the F-test implies *no selection*. We can assess in a separate test whether proportional selection is present: the null hypothesis $\hat{\omega}_{C1} = 1$ implies that $\theta_{1C} = \theta_{1\mathcal{T}}$, and thus it implies no proportional selection. If this null is not rejected in a t-test, then, given the rejection of the joint test, *level single selection* is present. If the same test (the null $\hat{\omega}_{C1} = 1$) is rejected, this implies that $\theta_{1C} \neq \theta_{1\mathcal{T}}$, and thus, proportional selection is present. We are then in one of

⁶Note that it is equivalent to specify the regression at the group level directly.

two possible remaining scenarios: there is *proportional single selection* or *double selection*. Our estimator works in either of these cases, including the one of no selection.

When the presence of (double) selection is established, the next test is a *pre-test* which addresses the stability of the selection process over time in the pre-policy periods. As such, it serves to judge whether the assumption of a constant (double) selection bias, which is at the core of the second step of the estimation, is warranted. This pre-test of *constant (double) selection* tests if the selection parameters, θ_{0g} and θ_{1g} , remain constant over time. Given the mapping between θ_{0g}, θ_{1g} and ω_{0g}, ω_{1g} , constant selection implies constant “quality” of the normalization across pre-policy periods. This equivalence can be expressed as: $\mathbb{E}_i[\tilde{q}_{igt}|\mathcal{C}] = \mathbb{E}_i[q_{igt}|\mathcal{T}]$, for $t < t_p$. To test this, consider the pre-policy treatment effects, $\hat{\gamma}_t$ for $t < t_p - 1$, using (22). If the selection bias is constant, these effects are statistically equal to zero, indicating no residual differences between the treated and normalized control groups in the pre-policy period.

Recent research has documented that the standard tests for pre-trends in the DID analyses tend to suffer from low power (Roth, 2022; Rambachan and Roth, 2023). The power of such tests increases with the number of pre-policy periods $t < t_p$. Likewise, the power of our pre-test for constant double selection will also increase with the number of pre-policy periods.

4.3 Inference

In the context of our two-step estimator, each step contains a challenge for inference. The challenge in the first-step regression of outcomes on outcomes, which delivers the estimates of $\hat{\omega}_0$ and $\hat{\omega}_1$, is that these parameters are identified solely through time variation in outcomes across groups. Standard inference procedures are not well-suited for estimating the standard errors in this context, because they overlook the fact that each observed group-level outcome represents an average across individuals within a time period and thus has its own sampling distribution. The additional challenge in the second-step estimation of the policy effect $\hat{\gamma}_t$ itself is that it needs to take into account the sampling uncertainty inherited from the first-step estimation in the predicted normalized outcomes $\hat{q}_{igt}$. To address this double challenge, we consider a non-parametric bootstrapping procedure.

Our proposed bootstrap procedure consists of performing the following steps M times: (i) construct bootstrap sample m by resampling with replacement from the observed outcomes q_{igt} within each time period and group, denote the bootstrap sample specific outcomes by q_{igt}^m ,⁷ (ii)

⁷For panel data, we recommend an alternative resampling strategy: rather than resampling within periods and groups, we suggest resampling entire individual histories. This form of block bootstrapping better preserves the dependence structure in the data as it accounts for the within-individual autocorrelation structure.

compute the average outcome for each group and time, $q_{gt}^m = \frac{\sum_i q_{igt}^m}{n_{gt}^m}$, where n_{gt}^m is the number of individuals in group g at time t ; (iii) estimate ω_0^m and ω_1^m ; then normalize the outcome for the control group following the procedure outlined in (19); and (iv) use the normalized outcomes to estimate the policy effect, γ_t^m . Note that (ii)–(iv) simply describe the estimation of the two-step estimator for each bootstrap sample. The estimates $\hat{\omega}_0^m$, $\hat{\omega}_1^m$, and $\hat{\gamma}_t^m$ are stored. The (joint) distribution of $\hat{\omega}^m$ then forms the foundation for the selection tests described in the previous section. Likewise, the distribution of $\hat{\gamma}_t^m$ serves to perform the pre-test and to assess significance of the estimated policy effects.

Alternatively, we can compute standard errors non-parametrically for $\hat{\omega}_0^m$, $\hat{\omega}_1$ and $\hat{\gamma}_t$ directly from (18), as we state in Proposition 5. This combines pre and post-policy variation between groups and across groups over time. Similarly, we can get standard errors for the policy effect $\hat{\gamma}_t$, as we state in Proposition 6.

Proposition 5. *Let (18) be estimable by OLS, then $\hat{\omega}_0$, $\hat{\omega}_1$ are unbiased with the following mean and variance:*

$$\text{Mean: } \mathbb{E}[\hat{\omega}_0] = \omega_0 \quad \text{Variance: } \text{Var}[\hat{\omega}_0] = \frac{\hat{\sigma}_\epsilon^2}{T} \frac{\sum_t (\hat{q}_{C,t}^2)}{\sum_t (\hat{q}_{C,t} - \hat{q}_C)^2} \quad (23)$$

$$\text{Mean: } \mathbb{E}[\hat{\omega}_1] = \omega_1 \quad \text{Variance: } \text{Var}[\hat{\omega}_1] = \frac{\hat{\sigma}_{\epsilon_T}^2 + 2(\hat{\sigma}_{\epsilon_C}^2 + \hat{\omega}_1^2)}{\sum_t (\hat{q}_{C,t} - \hat{q}_C)^2 - 2(\hat{\sigma}_{\epsilon_C}^2 + \hat{\epsilon}_C^2)} \quad (24)$$

With $\hat{q}_{gt} = \sum_{i \in g} q_{it} / n_{gt}$, $\hat{q}_g = \sum_t \hat{q}_{gt} / T$, $\hat{\epsilon}_{gt} = \sum_{i \in g} \hat{\epsilon}_{it} / n_{gt}$ and $\hat{\epsilon}_C = \sum_t \hat{\epsilon}_{g,t} / T$

Proof: see Appendix C.1.

Proposition 6. *Given $\hat{\omega}_0$, $\hat{\omega}_1$ with mean and variance as per (23) and (24), then $\hat{\gamma}_{T,t}$ has mean and variance given as follows:*

$$\text{Mean: } \mathbb{E}[\hat{\gamma}_{it}] = \gamma_t$$

$$\text{Variance: } \text{Var}[\hat{\gamma}_{it}] = \hat{\sigma}_{\epsilon,T}^2 + \hat{\sigma}_\epsilon^2 \left(1 + \frac{1}{T} + \frac{(\hat{q}_{C,t} - \hat{q}_C)^2}{\sum_t (\hat{q}_{C,t} - \hat{q}_C)^2} \right) + 2 \text{cov}(\hat{q}_{T,t}, \hat{q}_{C,t})$$

Proof: see Appendix C.2.

5 Empirical Illustration: Revisiting Card (1992)

We now revisit a case study of Card (1992), who explores the effect of an increase in the minimum wage in California in July 1988 on wages of different population groups. Our emphasis

is on illustrating our Double-Selection framework and to compare it with modern DiD approaches. Thus, we do not explore the different groups and possible outcome variables at the same level of detail as Card in his original study. Instead, we take a more birds-eye perspective and focus on the average wage at the state level for workers in the retail trade sector.

We first use the same control group as Card does in his original study, and show how the Double Selection estimator works in this context. We then consider all US states as possible controls for California one-by-one. Importantly, there are states which do not pass a traditional pre-test in levels or logs: based on those tests, a practitioner would deem a DiD approach as not viable to estimate the policy effect. Many of the same states, however, do provide a valid counterfactual through the lens of the Double Selection framework: The first-stage estimates uncover double selection, which in the second stage delivers a policy effect estimate based on the identifying assumption of constant double selection bias.

The Card (1992) Control. In order to construct a counterfactual for the evolution of average wages for some group in absence of the minimum wage increase, Card relies on the assumption of parallel trends of wages of that group in California and wages of the same group in a set of control regions that did not experience such policy change around the same time. We now use data from the Current Population Study (in particular, the outgoing rotation groups) from 1982–2024. For direct comparability, we first construct a control group resembling the one used by Card for which we construct average wages.⁸ Importantly, we use time series from before the policy change to implement the first stage of our Double Selection estimator. Similarly, for the DiD estimators, we use the same pre-policy time series in order to conduct pre-tests. It is noteworthy that Card does not pursue any pre-test: Instead, he argues for the validity of the parallel trends assumption by documenting that the samples of California and the Card Control are similar along a set of observable characteristics. Card then compares outcomes directly before (year 1987) and after (year 1989) the policy change (of 1987).

In panels (a1)–(a4) of Figure 5, we report the results of employing the Double Selection estimator using the set of states which Card uses to construct the counterfactual. Panel (a1) shows the time series of average hourly wages in California and this “Card Control” for 1982–94. In panel (a2), we move to the outcome-on-outcome space and plot California against the Card Control using data before the policy change, i.e., each marker shows the outcomes for California and the Card Control in one of the seven years from 1982–88. We also plot the fitted line from the first stage regression and report the OLS estimates and standard errors. The slope coefficient

⁸We consider a weighted average of people living in Arizona, Florida, Georgia, New Mexico, and Dallas-Fort Worth.

is $\hat{\omega}_1^{Card} = 0.82$ and with a standard error of 0.17 not statistically different from 1. The intercept coefficient is $\hat{\omega}_0^{Card} = 2.12$ (with a standard error of 1.02), and thus statistically significant. This implies that through the lens of our double selection framework, between California and the Card Control there is selection bias in levels.

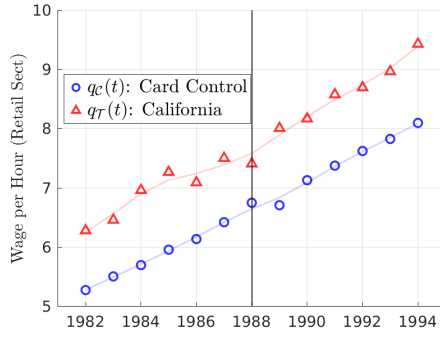
In panel (a3), we move to step 2 of the estimator and plot the time series of average wages for the two states, whereby we adjust the wages in the Card Control with the estimated $\hat{\omega}_0^{Card}$ and $\hat{\omega}_1^{Card}$, as derived in (19): $\hat{q}_{C,t} = \hat{\omega}_{C0} + \hat{\omega}_{C1}\mathbb{E}_i[q_{igt}|\mathcal{C}]$. In panel (a4), we show the difference between California and the adjusted Card Control, which delivers the estimated policy effect for each year. This policy effect is (statistically) flat at zero before the policy, and then turns positive for the years after the policy. The non-significance during the pre-policy time period lends confidence into the assumption of constant selection bias in line with the estimated one (see our corresponding discussion in Section 4.2). In the original Card (1992) paper, the DiD exercise is done on log wages (recall that it is done without a pre-test that resorts to the pre-policy time series): He finds that one year after the policy, average log wages increase by 0.05 (s.e. 0.02) in addition to the no policy comparison group. When we translate our double selection estimates into proportional change, those line up with the ones found in Card (1992): on average, the wage increase induced by the minimum wage hike is about 5 percent. Tracking the outcomes for some years reveals a growing difference between California and the Card control.

Widening the Scope. In panels (b1)–(b4), we redo the exact same analysis, but for a different state as the control group for California. In particular, we illustrate double selection using New Hampshire. The estimates of the outcome-on-outcome model in panel (b2) indicate strong proportional selection, as visible by a slope that is significantly different from 1: $\hat{\omega}_1^{New\ Hampshire} = 0.50$ (0.11). Further, the intercept is statistically different from 0. With both components of selection active, we move to the adjusted wages over time in panel (b3), which reveals that before policy, the double selection adjustment lines up the outcomes well, which is also confirmed by a zero policy effect in panel (b4). Finally, after policy, we find the policy effect estimated in the second step of the estimation. Interestingly, the estimate based on this alternative control group is broadly similar to the one that uses the Card Control. The most important insight is that we are in a setting, where a traditional DiD would not be applicable: it can control for either level or proportional selection, but not for both simultaneously. In such a setting, allowing for double selection enables the practitioner to pursue the analysis and estimate a policy effect based on a valid counterfactual.

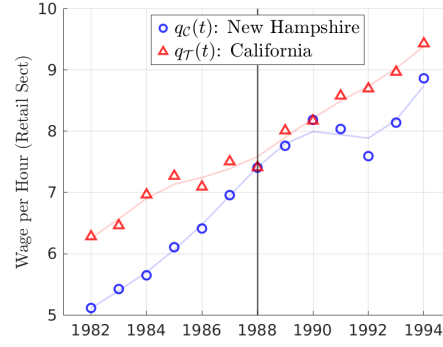
Motivated by this, we now explore a range of possible counterfactuals for California based on the set of US states. In Table 3, we report the number of states for which we find the different

Figure 5: Empirical Illustration: replicating Card (1992)

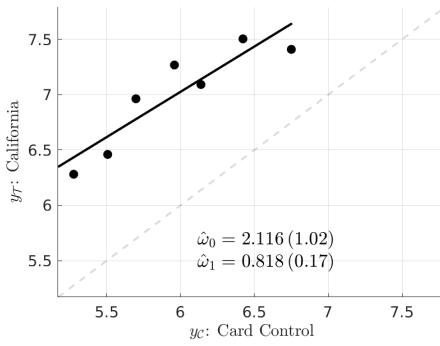
(a1) California vs. Card Control
Before Adjustment



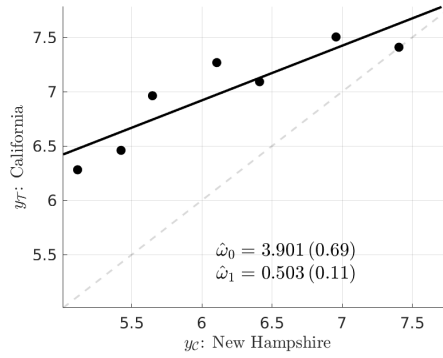
(b1) California vs. New Hampshire as Control
Before Adjustment



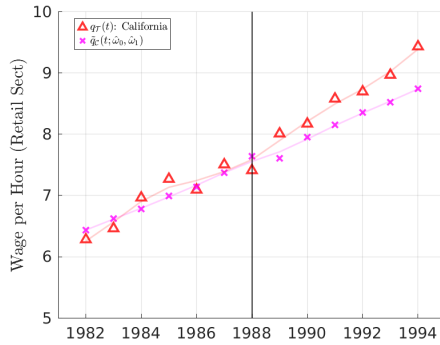
(a2) Outcome on Outcome Plot



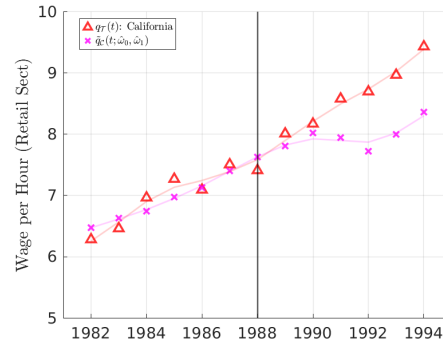
(b2) Outcome on Outcome Plot



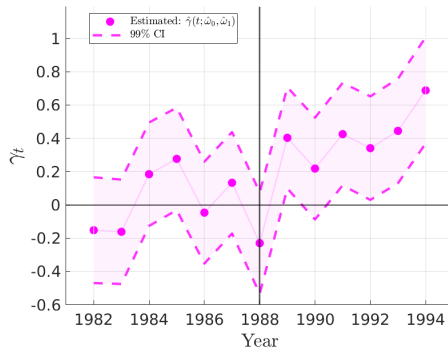
(a3) After Adjustment



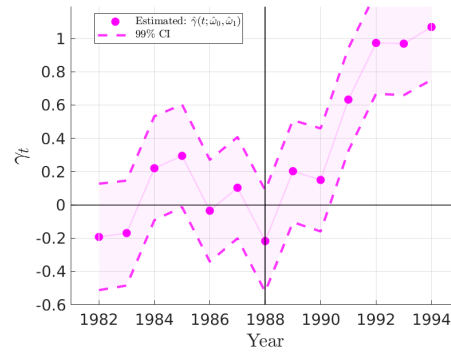
(b3) After Adjustment



(a4) Policy Eff.



(b4) Policy Eff.



Notes: See text.

Table 3: California 1988—Single and Double Selection Across US States

Selection Type	Frequency	Share	$\bar{\gamma}_{t_p+1}$ USD	$\bar{\gamma}_{t_p+1}/\tilde{y}_{C(t_p+1)}(\%)$
Single Selection: CSB Levels	10	20%	0.23	3.29%
Double Selection	13	26%	0.36	4.92%
Other	27	54%	-	-

Notes: Fitting state-by-state to California pre 1988, we consider cases with single selection in levels ($\omega_0 \neq 0$; $\omega_1 = 1$) or double selection ($\omega_0 \neq 0$ or $\omega_0 = 0$; $\omega_1 \neq 1$).

types of selection when fitting our first-stage estimator. For each of those we then estimate the effect of increasing the minimum wage in 1988 on averages wages in the year after the policy, 1989. We also translate the estimated level effect into proportional increases. In the table, we report the average over the states covered in each row. We find first-stage evidence for single selection in levels in 10 cases. In 13 cases, we find evidence for double selection. Thus, allowing for Double Selection more than doubles the number of states which through the lens of our framework can serve as control group—relative to the number states for which a DiD estimator is valid.

We further illustrate this by employing off-the-shelf DiD estimators. For each of the US states as potential control for California, we first run a formal pre-test for a DiD-in-levels estimator. For those that pass the pre-test, we estimate the policy effect, and report the mean estimate of those policy effects (as well as the corresponding standard deviation). We then do the same for the DiD-in-logs estimator, and finally our Double Selection estimator. Table 4 summarizes these results. [TBC]

6 Conclusion

We began this paper with a quote from [Angrist and Pischke \(2009\)](#) which emphasizes the incompatibility of two types of *constant* selection bias, one in levels and one in logs. Our work challenges this notion by proposing a unified constant *double*-selection framework in which both types of selection bias are present concurrently. In the DGP at the core of our framework, two group-specific shifters of a common time trend induce additive selection together with proportional selection. This leads to the common time trend not necessarily manifesting itself as a parallel trend over time in either levels or logs. We show that the relative dominance of the two types of selection depends on the magnitude of the common component and on the size of the two selection terms. We establish identification of the causal effect of a binary treatment within this framework. Key to identification is the assumption of constant double selection bias, which

is a modification of the standard constant selection bias assumption. The two standard cases—constant selection in levels or logs—are covered as special cases, which arise under restrictions of our more general DGP. After establishing identification, we propose an estimator in levels that is applicable regardless of whether the selection bias is additive or proportional. The estimator builds around the stable linear relationship of outcomes across groups: in the presence of both additive and proportional selection, the expected outcome of the treatment group is a linear transformation of the expected outcome of the control group. Moreover, because our estimator operates on outcomes in levels, it is suitable even in the presence of non-positive values.

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Appendix

A Notation table

Symbol	Definition
q_{igt}	Observed outcome for individual i in group g at time t
y_{igt}	Counterfactual (pre-policy) outcome, i.e., q_{igt} for $t < t_p$
λ_t	Common shock at time t
θ_{0g}	Group-specific level shifter
θ_{1g}	Group-specific proportional (log) shifter
ϵ_{igt}	Idiosyncratic shock, $E[\epsilon_{igt}] = 0$, $\text{Var}(\epsilon_{igt}) = \sigma_g^2$
u_{igt}	Rescaled error: $1 + \frac{\epsilon_{igt}}{\theta_{0g} + \theta_{1g}\lambda_t}$
g^*	Reference group (baseline)
$E[y_{igt}]$	Expected pre-policy outcome in levels
$E[\ln y_{igt}]$	Expected pre-policy outcome in logs
SB in levels	$E[y_{igt}] - E[y_{ig^*t}]$
SB in logs	$E[\ln y_{igt}] - E[\ln y_{ig^*t}]$
$\Theta_0 C_0$	Set of θ_0 vectors satisfying constraint C_0 : $\theta_{0g} = \theta_{0g^*} = 0$
$\Theta_1 C_1$	Set of θ_1 vectors satisfying constraint C_1 : $\theta_{1g} = \theta_{1g^*}$
$\bar{\lambda}$	Threshold value at which dominance between selection types switches

B Appendix for Section 2

B.1 Proof of Lemma 1

Proof. We prove (a), (b), and (c) separately:

(a) Since $y_{igt} \in \mathbb{R}^{++}$, we have $\mathbb{E}_i[y_{igt}] > 0$. By taking expectations, we obtain:

$$\mathbb{E}_i[y_{igt}] = \theta_{0g} + \theta_{1g}\lambda_t + \mathbb{E}_i[\epsilon_{igt}] = \theta_{0g} + \theta_{1g}\lambda_t > 0, \quad \forall(g, t).$$

Hence, the claim follows.

(b) Isolating u_{igt} from (2) and substituting y_{igt} from (1), we obtain:

$$\begin{aligned} u_{igt} &= \frac{y_{igt}}{\theta_{0g} + \theta_{1g}\lambda_t} \\ &= \frac{\theta_{0g} + \theta_{1g}\lambda_t + \varepsilon_{igt}}{\theta_{0g} + \theta_{1g}\lambda_t} = 1 + \frac{\varepsilon_{igt}}{\theta_{0g} + \theta_{1g}\lambda_t}. \end{aligned}$$

Since $y_{igt} \in \mathbb{R}^{++}$ and, by Lemma 1(a), $\theta_{0g} + \theta_{1g}\lambda_t > 0$, it follows that $u_{igt} > 0$, i.e.

$$u_{igt} - 1 = \frac{\varepsilon_{igt}}{\theta_{0g} + \theta_{1g}\lambda_t} > -1$$

Furthermore, since ε_{igt} has constant zero mean, the expected value of u_{igt} is:

$$\mathbb{E}_i[u_{igt}] = 1 + \frac{\mathbb{E}_i[\varepsilon_{igt}]}{\theta_{0g} + \theta_{1g}\lambda_t} = 1.$$

(c) Let

$$\ln(u_{igt}) = \ln(1 + (u_{igt} - 1)).$$

Since $\mathbb{E}_i[u_{igt}] = 1$ (Lemma 1(b)), we Taylor expand $\ln(u_{igt})$ around $u_{igt} = 1$ (e.g. Feller, 1971):⁹

$$\ln(u_{igt}) = (u_{igt} - 1) - \frac{(u_{igt} - 1)^2}{2} + \frac{(u_{igt} - 1)^3}{3} - \dots, \quad \text{for } |u_{igt} - 1| < 1.$$

Taking expectations on both sides:

$$\mathbb{E}_i[\ln(u_{igt})] = \mathbb{E}_i[u_{igt}(t) - 1] - \frac{1}{2}\mathbb{E}_i[(u_{igt} - 1)^2] + \frac{1}{3}\mathbb{E}_i[(u_{igt} - 1)^3] - \dots.$$

For small values of $u_{igt} - 1$, the second-order and higher-order terms (i.e., $\mathbb{E}_i[(u_{igt} - 1)^2]$, $\mathbb{E}_i[(u_{igt} - 1)^3]$, etc.) become negligible.¹⁰ Thus, the first-order approximation is:

$$\mathbb{E}_i[\ln(u_{igt})] \approx \mathbb{E}_i[u_{igt} - 1].$$

⁹To satisfy $|u_{igt} - 1| < 1$, it suffices to assume symmetry for ε_{igt} . Note that $\frac{\varepsilon_{igt}}{\theta_{0g} + \theta_{1g}\lambda_t}$ ($= u_{igt} - 1$) has mean zero and is bounded from below to be larger than minus 1 (Lemma 1(b)) and, hence, negative values of ε_{igt} satisfy $|u_{igt} - 1| < 1$. Thus, if ε_{igt} is symmetric then positive values of ε_{igt} also satisfy $|u_{igt} - 1| < 1$.

¹⁰The same argument is implicitly used to approximate growth rates with log differences when the growth rate r is small. Precisely, let $\ln y_{ig(t+1)} - \ln y_{igt} = \ln y_{igt}(1 + r_{igt}) - \ln y_{igt} = \ln(1 + r_{igt}) \approx r_{igt}$, thus: $\mathbb{E}_i[\ln y_{ig(t+1)} - \ln y_{igt}] = \mathbb{E}_i[\ln(1 + r_{igt})] \approx \mathbb{E}_i[r_{igt}]$.

□

Lemma B.1. *If the common component is constant, i.e. $\lambda_t = \lambda < \infty \ \forall t$, then the DGP in (1) exhibits constant selection bias (CSB) in both levels (3) and logs (4).*

B.2 Proof of Proposition 1.

Proof. We prove (a) and (b) separately:

(a) Plug (C1) into (3) and (4), respectively:

$$\text{Levels: } \mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}] = \theta_{0g} - \theta_{0g^*} \quad (\text{constant}) \quad (25)$$

$$\begin{aligned} \text{Logs: } \mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}] \\ = \ln(\theta_{0g} + \theta_{1g^*}\lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*}\lambda_t) \end{aligned} \quad (\text{not constant}) \quad (26)$$

Thus, the DGP displays a CSB in levels, see Equation (25). The log counterpart of the CSB in levels—i.e., the result of imposing on the selection bias in logs in Equation (4) the assumption (C1) that leads to the CSB in levels—is generally not constant (26), except if λ_t is constant.

(b) Plug (C0) into (3) and (4), respectively:

$$\text{Levels: } \mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}] = (\theta_{1g} - \theta_{1g^*})\lambda_t \quad (\text{not constant}) \quad (27)$$

$$\text{Logs: } \mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}] = \ln \theta_{1g} - \ln \theta_{1g^*} \quad (\text{constant}) \quad (28)$$

Thus the DGP displays a CSB in logs, see Equation (28). The level counterpart of the CSB in logs—i.e., the result of imposing on the selection bias in levels in Equation (3) the assumption (C0) that leads to the CSB in logs—is generally not constant (27), except if λ_t is constant.

□

B.3 Proof of Proposition 2.

Proof. We prove (a) and (b) separately:

(a) The selection bias in levels (3) can be rewritten as:

$$\begin{aligned} \text{Levels: } \mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}] &= \underbrace{(\theta_{0g} - \theta_{0g^*})}_{= \text{CSB in Levels (25)}} + \underbrace{(\theta_{1g} - \theta_{1g^*})\lambda_t}_{= \text{Level counterpart of CSB in Logs (27)}} \quad (\text{not constant}) \quad (29) \end{aligned}$$

where the first and second term separately capture each of the two types of selection in the DGP. The first term, $(\theta_{0g} - \theta_{0g^*})$, captures the CSB in levels (25), while the second term, $(\theta_{1g} - \theta_{1g^*})\lambda_t$, is the level counterpart of the CSB in logs (27).

(b) The selection bias in logs (4) can be rewritten as:

$$\begin{aligned} \text{Logs: } \mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}] &= \underbrace{\ln(\theta_{0g} + \theta_{1g^*}\lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*}\lambda_t)}_{= \text{Log counterpart of CSB in Levels (26)}} + w(\theta_0, \theta_1, \lambda_t) \underbrace{(\ln \theta_{1g} - \ln \theta_{1g^*})}_{= \text{CSB in Logs (28)}} \\ &\quad (\text{not constant}) \quad (30) \end{aligned}$$

where the first term is the log counterpart of the CSB in levels (26), while the second term is the CSB in logs (28) adjusted with weight

$$w(\theta_0, \theta_1, \lambda_t) = w(\theta_{0g}, \theta_{1g}, \theta_{1g^*}, \lambda_t) = \frac{\ln(\theta_{0g} + \theta_{1g}\lambda_t) - \ln(\theta_{0g} + \theta_{1g^*}\lambda_t)}{\ln \theta_{1g} - \ln \theta_{1g^*}}$$

□

B.4 Characterization of the Selection Bias

Lemmas B.2 and B.3 discuss the link between λ_t and the selection bias; Propositions B.1 and B.2 discuss dominance of log or level selection.

Lemma B.2. *Strict Monotonicity of the Selection Bias (SB) in λ_t :*

(a) *Single CSB in levels:* If $\theta_0 \notin \Theta_0 | \mathbf{C0}$ and $\theta_1 \in \Theta_1 | \mathbf{C1}$, then

(i) The SB in levels is not a function of λ_t .

(ii) The SB in logs is strictly monotonically decreasing (or increasing) in λ_t if $\theta_{0g} - \theta_{0g^*} > 0$ (or $\theta_{0g} - \theta_{0g^*} < 0$).

(b) *Single CSB in logs:* If $\theta_0 \in \Theta_0 | \mathbf{C0}$ and $\theta_1 \notin \Theta_1 | \mathbf{C1}$, then

(i) The SB in levels is strictly monotonically increasing (or decreasing) in λ_t if $\theta_{1g} - \theta_{1g^*} > 0$ (or $\theta_{1g} - \theta_{1g^*} < 0$).

(ii) The SB in logs is not a function of λ_t .

(c) **Double Selection:** If $\theta_0 \notin \Theta_0|\mathbf{C0}$ and $\theta_1 \notin \Theta_1|\mathbf{C1}$, then

(i) The SB in levels is strictly monotonically increasing (or decreasing) if $\theta_{1g} - \theta_{1g^*} > 0$ (or $\theta_{1g} - \theta_{1g^*} < 0$)

(ii) The SB in logs is strictly monotonically increasing (or decreasing) if $\frac{\theta_{1g}}{\theta_{1g^*}} > \frac{\theta_{0g}}{\theta_{0g^*}}$ (or $\frac{\theta_{1g}}{\theta_{1g^*}} < \frac{\theta_{0g}}{\theta_{0g^*}}$).

Proof. We separately prove (a), (b) and (c):

(a) Single CSB in levels: Take the first derivative of the SB in levels (25) and SB in logs (26) with respect to λ_t :

$$(i) \text{ Levels: } \frac{\partial (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \notin \Theta_0|\mathbf{C0}, \theta_1 \in \Theta_1|\mathbf{C1})}{\partial \lambda_t} = \frac{\partial (\theta_{0g} - \theta_{0g^*})}{\partial \lambda_t} = 0 \quad (31)$$

$$(ii) \text{ Logs: } \frac{\partial (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \notin \Theta_0|\mathbf{C0}, \theta_1 \in \Theta_1|\mathbf{C1})}{\partial \lambda_t} \\ = \frac{\partial (\ln(\theta_{0g} + \theta_{1g^*}\lambda_t) - \ln(\theta_{0g^*} + \theta_{1g}\lambda_t))}{\partial \lambda_t} \\ = \theta_{1g^*} \left(\frac{1}{\theta_{0g} + \theta_{1g^*}\lambda_t} - \frac{1}{\theta_{0g^*} + \theta_{1g}\lambda_t} \right) \neq 0 \quad (32)$$

where (32) is strictly lower (or greater) than zero if $\theta_{0g} - \theta_{0g^*} > 0$ (or $\theta_{0g} - \theta_{0g^*} < 0$). Thus, in the case of a single CSB in levels, the SB in levels is not a function of λ_t (31) and the SB in logs is strictly monotonically decreasing (or increasing) in λ_t if $\theta_{0g} - \theta_{0g^*} > 0$ (or $\theta_{0g} - \theta_{0g^*} < 0$) (see (32)).

(b) Single CSB in logs: Take the first derivative of the SB in levels (27) and SB in logs (28) with respect to λ_t :

$$(i) \text{ Levels: } \frac{\partial (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \in \Theta_0|\mathbf{C0}, \theta_1 \notin \Theta_1|\mathbf{C1})}{\partial \lambda_t} = \frac{\partial (\theta_{1g} - \theta_{1g^*})\lambda_t}{\partial \lambda_t} = \theta_{1g} - \theta_{1g^*} \neq 0 \quad (33)$$

$$(ii) \text{ Logs: } \frac{\partial (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \in \Theta_0|\mathbf{C0}, \theta_1 \notin \Theta_1|\mathbf{C1})}{\partial \lambda_t} = \frac{\partial (\ln \theta_{1g} - \ln \theta_{1g^*})}{\partial \lambda_t} = 0 \quad (34)$$

where (33) is strictly greater (or lower) than zero if $\theta_{1g} - \theta_{1g^*} > 0$ (or $\theta_{1g} - \theta_{1g^*} < 0$). Thus, in the case of a single CSB in logs, the SB in logs is not a function of λ_t (see (34)) and the SB in

levels is strictly monotonically increasing (or decreasing) in λ_t if $\theta_{1g} - \theta_{1g^*} > 0$ (or $\theta_{1g} - \theta_{1g^*} < 0$) (see (33)).

(c) Double Selection: Take the first derivative of the SB in levels (29) and SB in logs (30) with respect to λ_t :

$$(i) \text{ Levels: } \frac{\partial (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1})}{\lambda_t} = \theta_{1g} - \theta_{1g^*} \neq 0 \quad (35)$$

$$(ii) \text{ Logs: } \frac{\partial (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1})}{\partial \lambda_t} \\ = \theta_{1g^*} \left(\frac{1}{\theta_{0g} + \theta_{1g^*} \lambda_t} - \frac{1}{\theta_{0g^*} + \theta_{1g^*} \lambda_t} \right) \\ + \frac{\partial w(\theta_0, \theta_1, \lambda_t)}{\partial \lambda_t} (\ln(\theta_{1g}) - \ln(\theta_{1g^*})) \\ = \frac{\theta_{1g}}{\theta_{0g} + \theta_{1g} \lambda_t} - \frac{\theta_{1g^*}}{\theta_{0g^*} + \theta_{1g^*} \lambda_t} \quad (36)$$

where we use

$$\frac{\partial w(\theta_0, \theta_1, \lambda_t)}{\partial \lambda_t} = \frac{1}{\ln(\theta_{1g}) - \ln(\theta_{1g^*})} \left(\frac{\theta_{1g}}{\theta_{0g} + \theta_{1g} \lambda_t} - \frac{\theta_{1g^*}}{\theta_{0g^*} + \theta_{1g^*} \lambda_t} \right) \quad (37)$$

Then, (35) is strictly greater (or lower) than zero if $\theta_{1g} - \theta_{1g^*} > 0$ (or $\theta_{1g} - \theta_{1g^*} < 0$) and (36) is strictly greater (or lower) than zero if $\frac{\theta_{1g}}{\theta_{1g^*}} > \frac{\theta_{0g} + \theta_{1g} \lambda_t}{\theta_{0g^*} + \theta_{1g^*} \lambda_t}$ (or $\frac{\theta_{1g}}{\theta_{1g^*}} < \frac{\theta_{0g} + \theta_{1g} \lambda_t}{\theta_{0g^*} + \theta_{1g^*} \lambda_t}$). Thus, in the case of double selection, the SB in levels is strictly monotonically increasing (or decreasing) if $\theta_{1g} - \theta_{1g^*} > 0$ (or $\theta_{1g} - \theta_{1g^*} < 0$) (see (35)), and the SB in logs is strictly monotonically increasing (or decreasing) if $\frac{\theta_{1g}}{\theta_{1g^*}} > \frac{\theta_{0g}}{\theta_{0g^*}}$ (or $\frac{\theta_{1g}}{\theta_{1g^*}} < \frac{\theta_{0g}}{\theta_{0g^*}}$) ((36)). $\square$

Lemma B.3. Limiting Values of the Selection Bias:

(a) **Single CSB in levels:** If $\theta_0 \notin \Theta_0 | \mathbf{C0}$ and $\theta_1 \in \Theta_1 | \mathbf{C1}$, then the limiting values as λ_t approaches either zero or infinity for

(i) The SB in levels are:

$$\lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) = \lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) = \theta_{0g} - \theta_{0g^*} \neq 0 \quad (38)$$

(ii) The SB in logs are:

$$\lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) = \ln \theta_{0g} - \ln \theta_{0g^*} \neq 0 \quad (39)$$

$$\lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) = 0 \quad (40)$$

(b) Single CSB in logs: If $\theta_0 \in \Theta_0|\mathbf{C0}$ and $\theta_1 \notin \Theta_1|\mathbf{C1}$, then the limiting values as λ_t approaches either zero or infinity for

(i) The SB in levels are:

$$\lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) = 0 \quad (41)$$

$$\lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) = \begin{cases} \infty, & \text{if } \theta_{1g} - \theta_{1g^*} > 0 \\ -\infty, & \text{if } \theta_{1g} - \theta_{1g^*} < 0 \end{cases} \quad (42)$$

(ii) The SB in logs are:

$$\begin{aligned} \lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) &= \lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) = \\ &\ln \theta_{1g} - \ln \theta_{1g^*} \neq 0 \end{aligned} \quad (43)$$

(c) Double Selection: If $\theta_0 \notin \Theta_0|\mathbf{C0}$ and $\theta_1 \notin \Theta_1|\mathbf{C1}$, then the limiting values as λ_t approaches either zero or infinity for

(i) The SB in levels are:

$$\lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) = \theta_{0g} - \theta_{0g^*} \neq 0 \quad (44)$$

$$\lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) = \begin{cases} \infty, & \text{if } \theta_{1g} - \theta_{1g^*} > 0 \\ -\infty, & \text{if } \theta_{1g} - \theta_{1g^*} < 0 \end{cases} \quad (45)$$

(ii) The SB in logs are:

$$\lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) = \ln \theta_{0g} - \ln \theta_{0g^*} \neq 0 \quad (46)$$

$$\lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) = \ln \theta_{1g} - \ln \theta_{1g^*} \neq 0 \quad (47)$$

Proof. We prove (a), (b) and (c) separately,

(a) Single CSB in levels: Take the limiting values as λ_t approaches either zero or infinity of:

(i) The SB in levels (25):

$$\lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \in \Theta_1 | \mathbf{C1}) = \lim_{\lambda_t \rightarrow 0} (\theta_{0g} - \theta_{0g^*}) = \theta_{0g} - \theta_{0g^*} \neq 0 \quad (48)$$

$$\lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \in \Theta_1 | \mathbf{C1}) = \lim_{\lambda_t \rightarrow \infty} (\theta_{0g} - \theta_{0g^*}) = \theta_{0g} - \theta_{0g^*} \neq 0 \quad (49)$$

(ii) The SB in logs (26):

$$\begin{aligned} \lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \in \Theta_1 | \mathbf{C1}) &= \lim_{\lambda_t \rightarrow 0} (\ln(\theta_{0g} + \theta_{1g}\lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*}\lambda_t)) \\ &= \ln \theta_{0g} - \ln \theta_{0g^*} \neq 0 \end{aligned} \quad (50)$$

$$\begin{aligned} \lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \in \Theta_1 | \mathbf{C1}) &= \lim_{\lambda_t \rightarrow \infty} (\ln(\theta_{0g} + \theta_{1g}\lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*}\lambda_t)) \\ &= \lim_{\lambda_t \rightarrow \infty} \ln \left(\frac{\theta_{0g}}{\lambda_t} + \theta_{1g} \right) - \ln \left(\frac{\theta_{0g^*}}{\lambda_t} + \theta_{1g^*} \right) \\ &= \underbrace{\ln \theta_{1g}}_{=\ln \theta_{1g^*}} - \ln \theta_{1g^*} = 0 \end{aligned} \quad (51)$$

(b) Single CSB in logs: Take the limiting values as λ_t approaches either zero or infinity of:

(i) The SB in levels (27):

$$\lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \in \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) = \lim_{\lambda_t \rightarrow 0} (\theta_{1g} - \theta_{1g^*}) \lambda_t = 0 \quad (52)$$

$$\begin{aligned} \lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \in \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) &= \lim_{\lambda_t \rightarrow \infty} (\theta_{1g} - \theta_{1g^*}) \lambda_t \\ &= \begin{cases} \infty, & \text{if } \theta_{1g} - \theta_{1g^*} > 0 \\ -\infty, & \text{if } \theta_{1g} - \theta_{1g^*} < 0 \end{cases} \end{aligned} \quad (53)$$

(ii) The SB in logs (28):

$$\begin{aligned} \lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \in \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) &= \lim_{\lambda_t \rightarrow 0} (\ln(0 + \theta_{1g}\lambda_t) - \ln(0 + \theta_{1g^*}\lambda_t)) \\ &= \ln(\theta_{1g}) - \ln(\theta_{1g^*}) \neq 0 \end{aligned} \quad (54)$$

$$\begin{aligned} \lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \in \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) &= \lim_{\lambda_t \rightarrow \infty} (\ln(0 + \theta_{1g}\lambda_t) - \ln(0 + \theta_{1g^*}\lambda_t)) \\ &= \ln(\theta_{1g}) - \ln(\theta_{1g^*}) \neq 0 \end{aligned} \quad (55)$$

(c) Double Selection: Take the limiting values as λ_t approaches either zero or infinity of:

(i) The SB in levels (29):

$$\begin{aligned} \lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) &= \underbrace{\lim_{\lambda_t \rightarrow 0} (\theta_{0g} - \theta_{0g^*})}_{\substack{= \text{CSB in Levels} \\ \text{as } \lambda_t \rightarrow 0, \text{ Eq. (48)}}} + \underbrace{\lim_{\lambda_t \rightarrow 0} (\theta_{1g} - \theta_{1g^*}) \lambda_t}_{\substack{= \text{Level counterpart of CSB in logs} \\ \text{as } \lambda_t \rightarrow 0, \text{ Eq. (52)}}} \\ &= \theta_{0g} - \theta_{0g^*} \neq 0 \end{aligned} \quad (56)$$

$$\begin{aligned} \lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[y_{igt}] - \mathbb{E}_i[y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) &= \underbrace{\lim_{\lambda_t \rightarrow \infty} (\theta_{0g} - \theta_{0g^*})}_{\substack{= \text{CSB in Levels} \\ \text{as } \lambda_t \rightarrow \infty, \text{ Eq. (49)}}} + \underbrace{\lim_{\lambda_t \rightarrow \infty} (\theta_{1g} - \theta_{1g^*}) \lambda_t}_{\substack{= \text{Level counterpart of CSB in logs} \\ \text{as } \lambda_t \rightarrow \infty, \text{ Eq. (53)}}} \\ &= \begin{cases} (\theta_{0g} - \theta_{0g^*}) + \infty, & \text{if } \theta_{1g} - \theta_{1g^*} > 0 \\ (\theta_{0g} - \theta_{0g^*}) - \infty, & \text{if } \theta_{1g} - \theta_{1g^*} < 0 \end{cases} \\ &= \begin{cases} \infty, & \text{if } \theta_{1g} - \theta_{1g^*} > 0 \\ -\infty, & \text{if } \theta_{1g} - \theta_{1g^*} < 0 \end{cases} \end{aligned} \quad (57)$$

(ii) The SB in logs (30):

$$\begin{aligned} \lim_{\lambda_t \rightarrow 0} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) &= \\ \underbrace{\lim_{\lambda_t \rightarrow 0} (\ln(\theta_{0g} + \theta_{1g^*} \lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*} \lambda_t))}_{\substack{= \text{Log counterpart of CSB in Levels as } \lambda_t \rightarrow 0, \text{ Eq. (50)}}} &+ \underbrace{\left(\lim_{\lambda_t \rightarrow 0} w(\theta_0, \theta_1, \lambda_t) \right)}_{=0} \underbrace{\left(\lim_{\lambda_t \rightarrow 0} (\ln(\theta_{1g}) - \ln(\theta_{1g^*})) \right)}_{\substack{= \text{CSB in logs as } \lambda_t \rightarrow 0, \text{ Eq. (54)}}} \\ &= \ln(\theta_{0g}) - \ln(\theta_{0g^*}) + 0 * (\ln(\theta_{1g}) - \ln(\theta_{1g^*})) \\ &= \ln(\theta_{0g}) - \ln(\theta_{0g^*}) \neq 0 \end{aligned}$$

$$\begin{aligned} \lim_{\lambda_t \rightarrow \infty} (\mathbb{E}_i[\ln y_{igt}] - \mathbb{E}_i[\ln y_{ig^*t}]) | (\theta_0 \notin \Theta_0 | \mathbf{C0}, \theta_1 \notin \Theta_1 | \mathbf{C1}) &= \\ \underbrace{\lim_{\lambda_t \rightarrow \infty} (\ln(\theta_{0g} + \theta_{1g^*} \lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*} \lambda_t))}_{\substack{= \text{Log counterpart of CSB in Levels as } \lambda_t \rightarrow \infty, \text{ Eq. (51)}}} &+ \underbrace{\left(\lim_{\lambda_t \rightarrow \infty} w(\theta_0, \theta_1, \lambda_t) \right)}_{=1} \underbrace{\left(\lim_{\lambda_t \rightarrow \infty} (\ln(\theta_{1g}) - \ln(\theta_{1g^*})) \right)}_{\substack{= \text{CSB in logs as } \lambda_t \rightarrow \infty, \text{ Eq. (55)}}} \\ &= 0 + 1(\ln \theta_{1g} - \ln \theta_{1g^*}), \\ &= \ln \theta_{1g} - \ln \theta_{1g^*} \neq 0 \end{aligned} \quad (58)$$

where we use the fact that $w(\theta_0, \theta_1 \lambda_t)$ is :

$$\lim_{\lambda_t \rightarrow 0} w(\theta_0, \theta_1, \lambda_t) = \lim_{\lambda_t \rightarrow 0} \frac{\ln(\theta_{0g} + \theta_{1g} \lambda_t) - \ln(\theta_{0g} + \theta_{1g^*} \lambda_t)}{\ln \theta_{1g} - \ln \theta_{1g^*}} = \frac{\ln(\theta_{0g}) - \ln(\theta_{0g})}{\ln(\theta_{1g}) - \ln(\theta_{1g^*})} = 0 \quad (59)$$

and

$$\lim_{\lambda_t \rightarrow \infty} w(\theta_0, \theta_1, \lambda_t) = \lim_{\lambda_t \rightarrow \infty} \frac{\ln\left(\frac{\theta_{0g}}{\lambda_t} + \theta_{1g}\right) - \ln\left(\frac{\theta_{0g}}{\lambda_t} + \theta_{1g^*}\right)}{\ln \theta_{1g} - \ln \theta_{1g^*}} = 1$$

□

Having characterized the direction (Lemma B.2) and limiting behavior (Lemma B.3) of the selection bias as a function of the common shock, we now focus on the case of double selection, where both level and proportional components are present. The key question is: under what conditions does one type of selection dominate the other? Proposition 3 addresses this by identifying the threshold value of the common time trend λ_t at which the relative contribution of each component shifts.

Proposition B.1. Dominant Selection: *If $\theta_0 \notin \Theta_0 | \mathbf{C0}$ and $\theta_1 \notin \Theta_1 | \mathbf{C1}$, then*

- (i) **SB in Levels** *There is a unique threshold $\bar{\lambda}$, below which the SB in levels (29) is dominated by the CSB in logs (i.e. the level shifter differences across regions) as opposed to the CSB in logs (i.e. the proportional shifter differences across regions):*

$$\left| \underbrace{\theta_{0g} - \theta_{0g^*}}_{= \text{CSB in levels, Eq. (25)}} \right| > \left| \underbrace{(\theta_{1g} - \theta_{1g^*}) \lambda_t}_{= \text{Level counterpart of CSB in logs, Eq. (27)}} \right| \Leftrightarrow \lambda_t < \bar{\lambda} \quad (60)$$

and the opposite otherwise.

- (ii) **SB in Logs** *There is a unique threshold $\bar{\lambda}$, below which the SB in logs (30) is dominated by the CSB in levels (i.e. the level shifter differences across regions) as opposed to the CSB in logs (i.e. the proportional shifter differences across regions):*

$$\left| \underbrace{\ln(\theta_{0g} + \theta_{1g^*} \lambda_t) - \ln(\theta_{0g^*} + \theta_{1g^*} \lambda_t)}_{= \text{Log counterpart of CSB in levels, Eq. (26)}} \right| > \left| \underbrace{w(\theta_0, \theta_1, \lambda_t)}_{\in (0,1)} \left(\underbrace{\ln \theta_{1g} - \ln \theta_{1g^*}}_{= \text{CSB in logs, Eq. (28)}} \right) \right| \quad (61)$$

$$\Leftrightarrow \lambda_t < \bar{\lambda} \quad (62)$$

and the opposite otherwise.

Proof. The existence of the threshold $\bar{\lambda}$ in (60) and (62) follows from applying Lemma B.2 and Lemma B.3.

- (i) The LHS of (60) is constant, the RHS is strictly monotonic (increasing) in λ_t (Lemma B.2) and goes from 0, when λ_t approaches 0, to infinity, when λ_t approaches ∞ (Lemma B.3). Therefore there is a unique $\bar{\lambda}$ in which LHS crosses (equals) RHS.
- (ii) The RHS of (62) is strictly monotonic (increasing) in λ_t (Lemma B.2) bounded between 0, when λ_t approaches 0, and a constant, when λ_t approaches infinity (Lemma B.3), whereas the LHS of (62) is decreasing with λ_t (Lemma B.2), approaching a constant when λ_t approaches 0 and approaching 0 when λ_t approaches ∞ (Lemma B.3). Therefore there is a unique $\bar{\lambda}$ in which LHS crosses (equals) RHS.

□

Proposition B.2. Identity of the two thresholds

Proof. Set $\lambda_t = \bar{\lambda}$ in (60) and (62)

$$\text{From (60)} \quad \bar{\lambda} = \frac{|\theta_{0g} - \theta_{0g^*}|}{|\theta_{1g} - \theta_{1g^*}|} \tag{63}$$

$$\begin{aligned} \text{From (62)} \quad \left| \ln(\theta_{0g} + \theta_{1g^*} \bar{\lambda}) - \ln(\theta_{0g^*} + \theta_{1g^*} \bar{\lambda}) \right| &= \left| \ln(\theta_{0g} + \theta_{1g} \bar{\lambda}) - \ln(\theta_{0g} + \theta_{1g^*} \bar{\lambda}) \right| \\ \bar{\lambda} &= \frac{|\theta_{0g} - \theta_{0g^*}|}{|\theta_{1g} - \theta_{1g^*}|} \end{aligned} \tag{64}$$

This is the case because logs is monotonic transformation of levels, so the threshold is the same. □

Here are two alternative proofs.

Proof. The selection bias in logs (4) can be rewritten as:

$$\begin{aligned} \ln y_r(t) - \ln y_{r^*}(t) &= \mathcal{O}(\theta_0, \theta_1, \lambda_t) \underbrace{(\ln(\theta_{0g} + \theta_{1g^*} \lambda_t) - \ln(\theta_{0g^*} + \theta_{1r^*} \lambda_t))}_{\substack{=\text{Eq. (26), CSB in Levels} \\ (\text{specified in Logs})}} \\ &\quad + (1 - \mathcal{O}(\theta_0, \theta_1, \lambda_t)) \underbrace{(\ln(\theta_{1g}) - \ln(\theta_{1g^*}))}_{\substack{=\text{Eq. (28), CSB in Logs}}} \quad (\text{not constant}) \end{aligned} \quad (65)$$

with weights $\mathcal{O}(\theta_0, \theta_1, \lambda_t) = \frac{\ln\left(\frac{(\theta_{0g} + \theta_{1g} \lambda_t)}{(\theta_{0g^*} + \theta_{1g^*} \lambda_t)} \frac{\theta_{1r^*}}{\theta_{1g}}\right)}{\ln\left(\frac{(\theta_{0g} + \theta_{1g^*} \lambda_t)}{(\theta_{0g^*} + \theta_{1r^*} \lambda_t)} \frac{\theta_{1r^*}}{\theta_{1g}}\right)}$. The first term, $\ln(\theta_{0g} + \theta_{1g^*} \lambda_t) - \ln(\theta_{0g^*} + \theta_{1r^*} \lambda_t)$, captures the CSB in levels expressed in logs (Eq. (27)), and the second term, $\ln(\theta_{1g}) - \ln(\theta_{1g^*})$ is the CSB in logs (Eq. (28)). $\square$

Proof. The selection bias in logs (4) can be rewritten as:

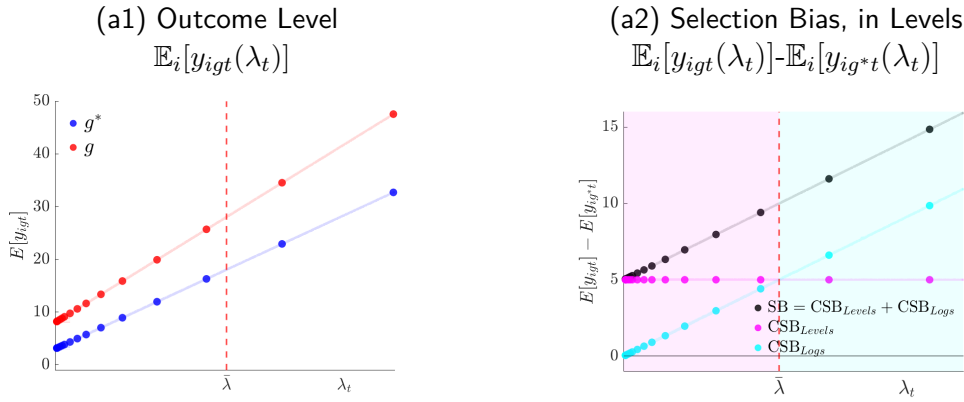
$$\begin{aligned} \ln y_r(t) - \ln y_{r^*}(t) &= \underbrace{\ln(\theta_{0g} + \theta_{1g^*} \lambda_t) - \ln(\theta_{0g^*} + \theta_{1r^*} \lambda_t)}_{\substack{=\text{Eq. (26), CSB in Levels} \\ (\text{specified in Logs})}} \\ &\quad + \underbrace{\ln(\theta_{1g}) - \ln(\theta_{1g^*})}_{\substack{=\text{Eq. (28), CSB in Logs}}} + \mathcal{O}(\theta_0, \theta_1, \lambda_t) \quad (\text{not constant}) \end{aligned} \quad (66)$$

with $\mathcal{O}(\theta_0, \theta_1, \lambda_t) = \ln\left(\frac{(\theta_{0g} + \theta_{1g} \lambda_t)}{(\theta_{0g^*} + \theta_{1g^*} \lambda_t)} \frac{\theta_{1r^*}}{\theta_{1g}}\right)$. The first term, $\ln(\theta_{0g} + \theta_{1g^*} \lambda_t) - \ln(\theta_{0g^*} + \theta_{1r^*} \lambda_t)$, captures the CSB in levels expressed in logs (Eq. (27)), and the second term, $\ln(\theta_{1g}) - \ln(\theta_{1g^*})$ is the CSB in logs (Eq. (28)). $\square$

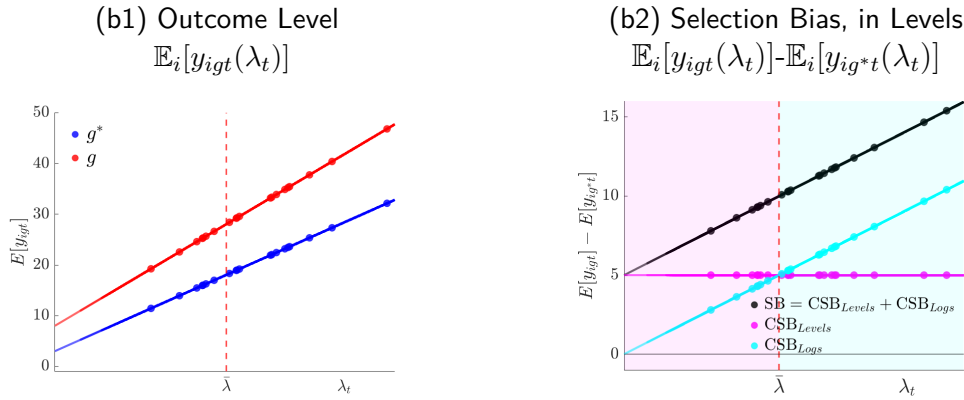
To complement the analytical results in this section, Figure B.1 plots the selection bias in levels and logs against the common component λ_t , holding group-specific parameters fixed. The figure shows that the relative contribution of the additive and proportional components to overall selection bias varies systematically with λ_t : At low values of λ_t , bias is primarily driven by level differences, while at high values, proportional differences dominate. The transition between these regimes occurs smoothly around a threshold $\bar{\lambda}$, consistent with the dominance conditions derived in Lemmas B.2 and B.3. Importantly, this relationship depends only on the magnitude of λ_t : By abstracting from time and plotting bias directly against λ_t , the figure highlights that the structure of selection bias under double selection depends fundamentally on the magnitude of the aggregate component, not on its time-series properties.

Figure B.1: Double Selection: Illustrating Alternative Common Time Trends λ_t

(a) A CTT with Deterministic Component: $\lambda_t = \lambda_0(1 + l_\lambda)^t + \eta_t$ and $\eta_t \sim iid(0, \sigma_\eta^2)$



(b) A Stochastic CTT: $\lambda_t = \bar{\lambda}e^{\eta_t}$ and $\eta_t \sim iid(0, \sigma_\eta^2)$



Notes: $\theta_{0g} > \theta_{0g^*}$, $\theta_{1g} > \theta_{1g^*} > 0$. Lines plot the deterministic component of the DGP; scattered dots plot the DGP including the stochastic component η_t . The axes are adjusted to ease the visualization of either CTT in levels or logs. (Parameters: $\theta_{0g^*} = 3$, $\theta_{1g^*} = 1.1$, $\theta_{0g} = 8$, $\theta_{1g} = 1.465$, $\bar{\lambda} = 13.69$; panel (a): $\lambda_0 = 3$, $l_\lambda = 0.001$, $\sigma_\eta^2 = 0.25$; panel (b): $\sigma_\eta^2 = 1.96$.)

C Appendix for Section 4

C.1 Proof of Proposition 5.

Proof. Let $\mathcal{C}$ and in $\mathcal{T}$ have n observations for every $t \leq t_p$ then:

$$\bar{q}_{\mathcal{T},t} = \theta_{\mathcal{T},0} + \theta_{\mathcal{T},1}\lambda_t + \sum_{i \in \mathcal{T}} \varepsilon_{i,t}/n_{\mathcal{T},t}$$

$$\bar{q}_{\mathcal{C},t} = \theta_{\mathcal{C},0} + \theta_{\mathcal{C},1}\lambda_t + \sum_{i \in \mathcal{C}} \varepsilon_{i,t}/n_{\mathcal{C},t}$$

Solve for λ_t

$$\lambda_t = (\bar{q}_{\mathcal{T},t} - \theta_{\mathcal{T},0} - \sum_{i \in \mathcal{T}} \varepsilon_{i,t}/n_{\mathcal{T},t})/\theta_{\mathcal{T},1} \quad (67)$$

$$\lambda_t = (\bar{q}_{\mathcal{C},t} - \theta_{\mathcal{C},0} - \sum_{i \in \mathcal{C}} \varepsilon_{i,t}/n_{\mathcal{C},t})/\theta_{\mathcal{C},1} \quad (68)$$

Equating (67) and (68) then solving for $\bar{q}_{\mathcal{T},t}$

$$\begin{aligned} \theta_{\mathcal{C},1}(\bar{q}_{\mathcal{T},t} - \theta_{\mathcal{T},0} - \sum_{i \in \mathcal{T}} \varepsilon_{i,t}/n_{\mathcal{T},t}) &= \theta_{\mathcal{T},1}(\bar{q}_{\mathcal{C},t} - \theta_{\mathcal{C},0} - \sum_{i \in \mathcal{C}} \varepsilon_{i,t}/n_{\mathcal{C},t}) \\ \bar{q}_{\mathcal{T},t} &= \frac{\theta_{\mathcal{T},1}}{\theta_{\mathcal{C},1}}(\bar{q}_{\mathcal{C},t} - \theta_{\mathcal{C},0} - \sum_{i \in \mathcal{C}} \varepsilon_{i,t}/n_{\mathcal{C},t}) + \theta_{\mathcal{T},0} + \sum_{i \in \mathcal{T}} \varepsilon_{i,t}/n_{\mathcal{T},t} \\ \bar{q}_{\mathcal{T},t} &= -\frac{\theta_{\mathcal{T},1}\theta_{\mathcal{C},0}}{\theta_{\mathcal{C},1}} + \theta_{\mathcal{T},0} + \frac{\theta_{\mathcal{T},1}}{\theta_{\mathcal{C},1}}\bar{q}_{\mathcal{C},t} + \sum_{i \in \mathcal{T}} \varepsilon_{i,t}/n_{\mathcal{T},t} - \frac{\theta_{\mathcal{T},1}}{\theta_{\mathcal{C},1}} \sum_{i \in \mathcal{C}} \varepsilon_{i,t}/n_{\mathcal{C},t} \\ \bar{q}_{\mathcal{T},t} &= \frac{\theta_{\mathcal{T},0}\theta_{\mathcal{C},1} - \theta_{\mathcal{T},1}\theta_{\mathcal{C},0}}{\theta_{\mathcal{C},1}} + \frac{\theta_{\mathcal{T},1}}{\theta_{\mathcal{C},1}}\bar{q}_{\mathcal{C},t} + \sum_{i \in \mathcal{T}} \varepsilon_{i,t}/n_{\mathcal{T},t} - \frac{\theta_{\mathcal{T},1}}{\theta_{\mathcal{C},1}} \sum_{i \in \mathcal{C}} \varepsilon_{i,t}/n_{\mathcal{C},t} \\ \bar{q}_{\mathcal{T},t} &= \omega_0 + \omega_1\bar{q}_{\mathcal{C},t} + \sum_{i \in \mathcal{T}} \varepsilon_{i,t}/n_{\mathcal{T},t} - \omega_1 \sum_{i \in \mathcal{C}} \varepsilon_{i,t}/n_{\mathcal{C},t} \end{aligned} \quad (69)$$

We estimate (69) through OLS:

$$\bar{q}_{\mathcal{T}t} = \omega_0 + \omega_1\bar{q}_{\mathcal{C}t} + \epsilon_t \quad \text{with } \mathbb{E}[\epsilon] = 0 \text{ and } \text{Var}[\epsilon] = \sigma_\epsilon^2 \quad (70)$$

We know that ϵ is the combination of three random variables, $\hat{\omega}_1$, $\epsilon_{\mathcal{T},t}$ and $\epsilon_{\mathcal{C},t}$, as such its mean and variance are explicitly:

$$\mathbb{E}[\epsilon] = \underbrace{\mathbb{E}[\epsilon_{i,\mathcal{T},t}] + \mathbb{E}[\epsilon_{i,\mathcal{C},t}, \hat{\omega}_1]}_{\mathbb{E}[\epsilon_{i,\mathcal{T},t} + \epsilon_{i,\mathcal{C},t} \cdot \hat{\omega}_1] = 0} + \underbrace{\mathbb{E}[\epsilon_{i,\mathcal{C},t}] \cdot \mathbb{E}[\hat{\omega}_1]}_{\mathbb{E}[\epsilon_{i,\mathcal{C},t} \cdot \hat{\omega}_1] = 0} = 0 \quad (71)$$

$$\text{Var}[\epsilon] = \sigma_\epsilon^2 = \underbrace{\sigma_{\epsilon_{i,\mathcal{T}}}^2 + \text{Var}[\epsilon_{i,\mathcal{C}} \cdot \hat{\omega}_1]}_{\text{Var}[\epsilon_{i,\mathcal{T},t} + \epsilon_{i,\mathcal{C},t} \cdot \hat{\omega}_1]} + \underbrace{\sigma_{\epsilon_{i,\mathcal{C}}}^2 \cdot \mathbb{E}[\hat{\omega}_1]^2 + \sigma_{\hat{\omega}_1}^2 \cdot \mathbb{E}[\epsilon_{i,\mathcal{C},t}]^2 + \sigma_{\epsilon_{i,\mathcal{C}}}^2 \cdot \sigma_{\hat{\omega}_1}^2}_{\text{Var}[\epsilon_{i,\mathcal{C}} \cdot \hat{\omega}_1]} \quad (72)$$

Recall that $q_{i,t}$ is a random variable with group $\mathcal{G}$ population mean $\bar{q}_{g,t} = \mathbb{E}_i[q_{i,t}|\mathcal{G}]$, and variance $\sigma_{g,t}^2 = \text{Var}_i[q_{i,t}|\mathcal{G}]$ (notice we do not need to assume normality at this point)

Let $\hat{q}_{g,t} = \sum_i q_{i,g,t}/n_{g,t}$ be our estimate of the group population mean $\bar{q}_{g,t}$,

Let $\hat{\sigma}_{g,t}^2 = \sum_i (q_{i,g,t} - \hat{q}_{g,t})^2 / (n_{g,t} - 1)$ be our estimate of the population variance $\sigma_{g,t}^2$

Explicitly we estimate,

$$\hat{\omega}_0 = \hat{q}_{\mathcal{T}} - \hat{\omega}_1 \hat{q}_{\mathcal{C}} \quad \text{with} \quad \hat{q}_g = \sum_t q_{g,t} / T$$

$$\hat{\omega}_1 = \frac{\sum_t (\hat{q}_{\mathcal{T},t} - \hat{q}_{\mathcal{T}})(\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2}$$

$$\hat{\sigma}_\epsilon^2 = \frac{\sum_t \hat{\epsilon}_t^2}{T-2} \quad \text{with} \quad \hat{\epsilon}_t = \hat{q}_{\mathcal{T},t} - \hat{q}_{\mathcal{T}} = \frac{\sum_i q_{i,\mathcal{T},t}}{n_{\mathcal{T},t}} - \left(\hat{\omega}_0 + \hat{\omega}_1 \frac{\sum_i q_{i,\mathcal{C},t}}{n_{\mathcal{C},t}} \right)$$

With variance covariance matrix for $\hat{\omega} = [\hat{\omega}_0, \hat{\omega}_1]$

$$\widehat{\text{Cov}}(\hat{\omega}) = \begin{bmatrix} \frac{\hat{\sigma}_\epsilon^2}{T} \frac{\sum_t (\hat{q}_{\mathcal{C},t}^2)}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2} & -\frac{\hat{\sigma}_\epsilon^2}{T} \frac{\sum_t (\hat{q}_{\mathcal{C},t})}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2} \\ -\frac{\hat{\sigma}_\epsilon^2}{T} \frac{\sum_t (\hat{q}_{\mathcal{C},t})}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2} & \frac{\hat{\sigma}_\epsilon^2}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2} \end{bmatrix}$$

Standard errors for $\hat{\omega}_0$ and $\hat{\omega}_1$ cannot be easily obtained as usual (by taking the square root of the diagonal elements of the estimated covariance matrix). But by solving for the value of $\sigma_{\hat{\omega}_0}^2$ and $\sigma_{\hat{\omega}_1}^2$ from the known elements of the covariance matrix:

$$\hat{\sigma}_{\hat{\omega}_0}^2 = \frac{\hat{\sigma}_\epsilon^2}{T} \frac{\sum_t (\hat{q}_{\mathcal{C},t}^2)}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2}$$

$$\hat{\sigma}_{\hat{\omega}_1}^2 = \frac{\hat{\sigma}_\epsilon^2}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2}$$

Knowing that $\hat{\sigma}_\epsilon$ is a function of $\hat{\sigma}_{\hat{\omega}_1}^2$ as per (72) □

Further, the mean and variance for the predicted values $\hat{q}_{\mathcal{T},t}$ are given by:¹¹

$$\mathbb{E}(\hat{q}_{\mathcal{T},t}) = \hat{\omega}_0 + \hat{\omega}_1 \hat{q}_{\mathcal{C},t} \quad (73)$$

$$\text{Var}(\hat{q}_{\mathcal{T},t}) = \hat{\sigma}_\epsilon^2 \left(1 + \frac{1}{T} + \frac{(\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2} \right) \quad (74)$$

Where $1/T$ is the extra variance for adding one more point in $\mathcal{T}$, each extra point variance share is $1/T$. And $\frac{(\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2}$ is the extra variance for adding one more point $\hat{q}_{\mathcal{C},t}$, each extra point variance share is its respective variance share with respect to the total variance of $\hat{q}_{\mathcal{C},t}$ over all T .

C.2 Proof of Proposition 6.

Proof.

- (1) Recall that the outcome is $q_{i,g,t} = \underbrace{\theta_{g0} + \theta_{g1}\lambda_t + P_{g,t}\gamma_{g,t}}_{q_{g,t}} + \varepsilon_{i,g,t}$ where $\varepsilon_{i,g,t} \sim iid(0, \sigma_g^2)$

As before, let $\hat{q}_{g,t} = \sum_i q_{i,g,t}/n_{g,t}$ be our estimate of the population mean $q_{g,t}$ as well as the group $\mathcal{G}$ population mean.

And, let $\hat{\sigma}_g^2 = \sum_i (q_{i,g,t} - \hat{q}_{g,t})^2 / (n_{g,t} - 1)$ be our estimate of the population variance σ_g^2 as well as the group $\mathcal{G}$ population mean. Alternatively we could also average out over all periods, $\hat{\sigma}_g^2 = \sum_t \sum_i (q_{i,g,t} - \hat{q}_{g,t})^2 / T(n_{g,t} - 1)$ ¹²

- (2) From 73 and 74 we know that the mean and variance for normalized outcome for the post policy period $\hat{q}_{\mathcal{C},t}$ are

$$\begin{aligned} \mathbb{E}[\hat{q}_{\mathcal{C},t}] &= \mathbb{E}[\hat{\omega}_0 + \hat{\omega}_1 \hat{q}_{\mathcal{C},t}] = \mathbb{E}[\hat{q}_{\mathcal{T},t}] - \gamma_t \\ \text{Var}[\hat{q}_{\mathcal{C},t}] &= \hat{\sigma}_\epsilon^2 \left(1 + \frac{1}{T} + \frac{(\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2}{\sum_t (\hat{q}_{\mathcal{C},t} - \hat{q}_{\mathcal{C}})^2} \right) \end{aligned}$$

- (3) Finally $\gamma_{\mathcal{T},t} \forall t$ is the sum of two, non-independent, random variables:

$$\begin{aligned} \hat{\gamma}_{\mathcal{T},t} &= \hat{q}_{\mathcal{T},t} - \hat{q}_{\mathcal{C},t}, \text{ with:} \\ \mathbb{E}[\hat{\gamma}_{\mathcal{T},t}] &= \mathbb{E}[q_{\mathcal{T},t} - \hat{q}_{\mathcal{C},t}] = \mathbb{E}[q_{\mathcal{T},t}] - \mathbb{E}[\hat{q}_{\mathcal{T},t}] + \gamma_t = \gamma_t \\ \text{Var}[\hat{\gamma}_{\mathcal{T},t}] &= \hat{\sigma}_{q_{\mathcal{T},t}}^2 + \hat{\sigma}_{\hat{q}_{\mathcal{C},t}}^2 + 2 \text{cov}(\hat{q}_{\mathcal{T},t}, \hat{q}_{\mathcal{C},t}) \end{aligned}$$

□

¹¹See William Greene, Chapter 4, Section 4.8.1 Prediction Intervals for the derivation for the multivariate case.

¹²At this step we also test whether σ_g^2 is constant over time.